Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 — UNIT 3 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_03 · UNIT 3

1 UE26ME142A · Unit_03: UNIT 3

This unit master organizes source material for Engineering Mechanics 1, Unit_03, officially titled UNIT 3. It includes 12 video transcripts and 14 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26ME142A · Unit_03: UNIT 3

Official catalogue entry: UE26ME142A, Unit_03 (UNIT 3). Bucket folder present: yes. Videos: 12; slide decks: 14.

EV0000001

2 CLASS 33 Introduction to Centroid

Study focus: CLASS 33 Introduction to Centroid. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

If you consider any external force in nature, each and every external force is a distributed force because external forces are applied mechanically on a body and they will have a definite area of contact.

Key details

- However small the area might be.
- That's why each and every external force is a distributed force.
- Actually it's not a point, it's the area of the tyre which is in contact with the pavement.

- You can see in this enlarged weave.
- But actually, it is a distributed force as shown in this figure.
- We are going to assume the same force is a concentrated force because the area of the contact between the tire and the surface or the pavement is very, very small compared to the other dimensions of the car.
- It is a concentrated force by considering the resultant of this distributed force.
- These are the few examples of the distributed forces.
- And in actual sense, concentrated forces doesn't exist in nature.
- Please remember this point.
- Study about the earth's gravitational attraction.
- In this line, I am going to explain you how to locate the center of gravity of any three-dimensional body practically.
- Body, three dimensional body of any size and shape.
- Let M be the mass of this body.
- Through the point A, the body is suspended with the help of this card.
- If I draw the free body diagram of this body, there are only two external forces acting on the body.
- The first one is tension in the card, and the one more is the self-weight of this body.
- This body is in equilibrium.
- These two external forces, one is tension and one more is self-weight.
- These two should be equal, opposite, and collinear.
- That's why this W is along the same line as that of this tension in the card, and it is downward because the tension in the card is upward.
- This is the case when the body is suspended through point A.
- Again, we are going to draw the free body diagram of this body.
- There are only two external forces.
- One is tension in the card and one more is the self rate.
- The body is in equilibrium, so these two forces should be equal, opposite and collinear, then only they'll get cancel each other, then the body will be in a state of equilibrium.
- That's how we are going to get the line of action of this.

Example or application from the lecture

- I'm going to consider the example of this automobile wheel, which is in contact with the pavement as shown in this figure.
- If you remember the numericals of any automobile example, like a car, when we consider the free body diagram of a car, we are going to consider the reaction from the pavement onto the wheel like this, as a concentrated force.
- Example, when we consider the distance between the two wheels, the area of this contact is very small.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

W It will be along the same line as that of the card, but in the opposite direction.

Key details

- And this also we are going to mark by drilling a small hypothetical hole on this rigid body.
- If you observe the line of action for the various cases, all the lines of action are concurrent at a particular point.
- That point we call it as the center of gravity.
- Irrespective of the orientation of the body this point will be remains the same.
- Then how to locate determine the center of gravity the same g mathematically.
- So to locate the center of gravity we are going to apply the concept of method of moments.
- Let g be the center of gravity of this body.
- Then the coordinates of this g will be always x bar, comma, y bar, comma, z bar.
- To find the value of X bar Y bar Z bar mathematically, we are going to apply the principle of moments.
- For this, what we are going to assume is we know each and every body will be having infinite number of.
- Each and every particle will be acted upon by the earth's gravitational force.
- Downward and the magnitude of that force is DW.
- Since the particle is very small,.
- It's just like a point, and the coordinates of this point are x, y, comma, z.
- According to this statement of principle of moments.
- The moment of the resulting gravitational force W about any axis it can be X, Y, R Z, will be always equal to.
- Some of the moments of the gravitational forces acting on all these particles about the same axis.
- This is the standard statement of the principle of moments.
- Then how to find the resultant gravitational force here.
- The resultant gravitational force W will be equal to sum of the weights of all these particles within the body.
- Next, moment of the.
- This particle about x axis.
- The moment of this particle about x axis will be equal to force into perpendicular distance.
- What is force?
- And distance is y.
- This is the moment of only one particle.
- Next we want the sum of the moments.
- Moment of the resulting gravitational force that is LHs is forces w into perpendicular distances y bar.
- Once we simplify we are going to get the value of y bar.
- Capital W And this W is total weight of the body.
- We got the value of this y bar.
- next we know the weight of the body will be always equals to mass into gravity.
- Substitute the value of w and dw here, we are going to get the one more expression for x bar, y bar and z bar.

- This is independent of the gravity.

Example or application from the lecture

- Here, let's consider there is a three-dimensional body of any size and shape.
- Let's apply here the moment of the resultant gravitational force W.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the point through which we assume the entire mass of the body is concentrated.

Key details

- This is called as center of mass.
- X bar, y bar, and z bar gives us the coordinates of the center of mass.
- The center of mass and center of gravity will coincide as long as the acceleration due to gravity is constant throughout the body, and at the same time, the forces are parallel to each other.
- Next, we know dense mass is equal to density into volume.
- And volume also we know area into thickness.
- Uniform density throughout as well as a uniform thickness throughout.
- In that case in both numerator and denominator, we are going to get.
- Density rho as constant as well as the thickness also as constant.
- coordinates of the centroid of the lamina.
- And centroid is the point where we assume the entire area of the body is concentrated.
- We are going this is the general formula to look at the centroid x bar integral of x into dA by total area, y bar integral of y into d a by total area.
- But when we are.
- considering any elemental strip within that lamina, then instead of considering the general expression, we are going to be more specific.
- if you look at this figure, there is a lamina something like this.
- Then I want to locate the centroid of this lamina.
- strip within this lamina.
- Then the coordinates of the centroid of this.
- Point R XC, Y C.
- The formula is going to be X bar is equals to integral of X C into DA by total area.
- Y bar is integral of.
- Yc into DA by total area.
- Next, there is one more formula.
- That's why this card is a first movement of.
- Next, there is one more formula.
- X bar is equals to summation of AI Xi by total area.
- Y bar is summation of AI Yi by total area.
- We are going to make use of.

- This particular formula in the composite areas.
- Composite area is nothing but any lamina figure which will be having some basic figures like combination of basic figures like.
- triangle, rectangle, square, circle, semicircle like that.
- And we can write it as a1 x1 plus a2 x2 plus so on.
- Plus a nxn by total area.
- That's how we are going to expand this particular formula.
- Next is axis of symmetry.
- You can look at this.
- Here there is a right circular cone.
- If you draw vertical line passing through the apex here through this point.
- So the center of gravity R centroid will be on this because.
- The moment of left of this will be equals to moment of right of this, these two will get cancel each other.
- It is applicable even for laminus as well.

Example or application from the lecture

- In that case, I am going to consider this elemental strip.
- We are going to make use of the concept of first moment of area to locate the center of a given lamina.
- Axis of symmetry is any line at the axis which divides the given lamina or a figure at any three-dimensional body into two equality.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

This is one more figure.

Key details

- If we draw a line passing through this, this right of the section is the mirror image of the left of the section.
- Then capital A is the area of this triangle.
- Let that be equals to A is equals to B into H.
- Next, I need to find the Y coordinate of the centrate.
- This is before that this is the reference axis, let it be x axis.
- The width, the depth of this element need to be dy.
- This element can be anywhere from this to this point T.
- And the depth of the element dy is also a variable.
- It can be.
- Anything from 0 to H.
- Next, the width of the element is also a variable because when the value of y is equals to 0, the width of this element will be equal to b.
- As the value of y increases, the width of the element decreases.

- That's why I'm going to call it as x.
- To find the y bar we have a formula y bar is equal to integral of yc into d a by total area e.
- DA is area of this element.
- The area of the element, since it is in the form of rectangle, almost like a rectangle, d A will be equal to B into H.
- Here B is X into depth of the element is DY.
- Y C we should find.
- The what is YC?
- This shaded area will have a centrite C.
- When Dy is very small, this Yc is approximately equal to Y.
- This will be equal to integrate integral of y.
- Yc is y into d is x into dy divided by total area capital E.
- X is a variable and y is also variable.
- Because we have the term dy here.
- This is x.
- And this we know b, and this we know y.
- And the total height of the rectangle is h.
- This is also known to us.
- We can use the concept of similar triangles because these two are parallel to each other.
- By using the concept of similar triangles,.
- Base of the smaller triangle that is x by base of the bigger triangle that is b will be equals to height of the smaller triangle.
- The height of the smaller triangle is only this.
- This will be equals to.
- Total h minus y.

Example or application from the lecture

- Next, let's find the y coordinate of central of a triangle of base B and a height H from the method of integration.
- You can consider either a right angle triangle or just any triangle, I'm going to consider a triangle like this.
- Let us consider a triangle A, B, C, Let B be the base of this triangle, let H be the height of the triangle.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

x in the form of constants b and h and a variable y.

Key details

- I'll take 1 by s here integral of.
- Y into X is B into H minus Y by H into D Y.
- This is 1 by A.

- As I said before, B and H are constants, so you can take them out.
- B and H.
- Y is varying from 0 to H.
- Then in Y into H is H into 1.
- Minus y square into d y d by outside the bracket.
- Simplify this, this will be equals to area is b into h by half into base into height.
- This is y cube by three zero two h.
- Next, this will be equals to.
- Two H to by B will get canceled here H square into.
- Once we substitute the value of y as h, so h square minus of h cube by 3, and now the next term is going to be 0.
- This is going to be H cube by six.
- This will be equal to two three.
- One by three H.
- This is the Y coordinate of the central of a triangle with respect to bottom plus member.
- This is x-axis passing to the bottom most member this is where the centroid will be, and this is y bar that is h by three.
- If the axis is passing to the topmost member, x-axis is passing to the topmost member at the apex of the triangle.
- In that case, the y-coordinate of the central is 2h by 3.
- That's it for today's class.

Example or application from the lecture

- This is a triangle if we consider.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

3 CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina

Study focus: CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Centred of an area of a circular sector.

Key details

- Look at this sector.
- This sector is symmetrical about x axis.
- The purpose for considering this sector symmetrical about x-axis is.
- The center of this sector will be an x-axis, so its y coordinate will be 0.
- In that case, the x coordinate of the centroid will be zero.
- We need to find only the value of y-coordinate of the centroid.
- That's the purpose of considering the figure this way.
- Next, let O be the center of this sector, let capital R be the radius of the sector, then the total angle of this sector, the inscribe the angle let it be 2 alpha.
- Since the total angle is 2 alpha, this angle with respect to x axis will be alpha and the remaining angle will also be alpha.
- Next we need to find only the y coordinate of the.
- Sorry, x coordinate of the centroid.
- To find the x coordinate of the centroid, we have a formula x bar is equal to integral of x into dA divided by total area.
- Which is varying along this radial line.
- It can be anywhere from 0 to capital R.
- Since it's a variable, I'm going to call it as.
- The distance of that element with respect to the center wall, let it be R0.
- Then the radial width is a variable, let it be DR0.
- Angle made by that element with respect to x axis, let it be theta.
- Then the inscriptor the angle of that element let it be d theta.
- D theta will be varying, it's a variable, and theta will be varying from minus alpha to.
- Plus alpha and R0 is varying from 0 to capital R.
- This shaded region is the area which we have considered for the using the formula x bar is equals to integral of x c into d divided by total area.
- Next, what is.
- XC and what is DA in this?
- DA is nothing but area of this element.
- This element is almost like a rectangle.
- What is B here?
- That is the width of the element that is equal to DR0.
- And what is H?
- H is nothing but the arc length that will be equal to R0 into inscribed angle D theta.
- Next, we need to find the value of x is c.
- X c is the centroidal distance of the element with respect to y axis, that x coordinate of the centroid of that smaller element with respect to.
- This is the x coordinate of that element, that itself is x.
- Then how to find this xc?
- When we need to find this XC, we are going to make one assumption that d theta is very small.
- When d theta is very small, theta plus d theta will be approximately equals to theta, and in that case, this element will be just like a.
- Then in that case, how to find X?

- That will be equal to R0 into cos theta.
- We are going to use the cosine rule there.
- Once we know the value of X C and DA, now we should know the value of capital A.
- Capital A is the area of the sector.
- Area of the sector will be equal to half into R squared into inscribed angle, that is to alpha, once we simplify that, we are going to get the value of A as R square into alpha.
- If you don't remember this formula, just integrate this BA.
- And R will be varying from 0 to capital R, and theta is varying from minus alpha to plus alpha.
- If we integrate this by applying the limits, then also we are going to get the value of area of sector.
- Once we know every term in the equation one.
- Substituting the equation one x is r naught cos theta, then d a is r naught d theta into dr0 by capital A.

Example or application from the lecture

- For this, I am going to consider a sector which is symmetrical about.
- And in one more way, you can consider a sector which will be symmetrical about.
- To apply this formula, we have to consider an element within this sector.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

This apply the limits as I said before R0 is varying from 0 to capital R and R theta is varying from minus alpha to plus alpha.

Key details

- Once we substitute the value of A, we are going to get the X coordinate of the centroid of the sector.
- That will be equal to 2r sin alpha by 3 alpha.
- And to use this formula, you have to remember the alpha should be in terms of.
- radiance and try to understand this formula in this way.
- The distance of the centroid with respect to center on the axis of symmetry that is 2 r sin alpha by.
- Alpha because always the centroid will be on the axis of symmetry for a sector from center to that point centroid on the axis of symmetry is 2 R sine alpha by 3 alpha.
- Again alpha will be radiance and r is the radius of the sector.
- If the semicircle is symmetrical about y-axis, then the x coordinate of the centroid will be 0 because the centroid will be on the axis of symmetry.
- That's the reason why I have considered the figure like this.
- A semicircle which is symmetrical about x-axis, then y-coordinate of the centroid will be 0.
- You need to find only x-coordinate.
- Y bar will be equals to integral of yc into da divided by total area.

- Then how to find yc and dA and total area?
- It can be anywhere from this zero to capital R.
- It is varying in this way.
- Let the inclination of that element with respect to the reference axis, that is x-axis, let it be theta.
- Then the inscribed angle let it be d theta.
- Then the width of the element is also a variable along radial direction.
- That's why we are going to call it as DR naught.
- Next, in this formula, DA stands for area of this element, that is shaded area.
- When d theta is comparatively small, this element will be almost like a rectangle.
- Here B is DR0, that is width of the element.
- Next, we need to find the depth of the element that will be equal to arc length, that is r0 into d theta.
- In this way we can find the value of dA.
- Next, what is yc?
- Yc is nothing but Y coordinate of the centroid of the element.
- In that case, the y distance of that element will be equal to r naught sin theta.
- Because here we are going to get a right angle triangle like this.
- That's why yc will be equal to R0 into sine theta.
- Then substitute the value of BA, then Y C in the equation 1.
- Then capital A is area of the semicircle that is I R square by 2.
- Once we substitute.
- in the equation one, we are going to get it as integral of r naught sin theta.
- into r0 d theta into dr0 divided by total area.
- As I said before, r naught is varying from 0 to capital R and theta is varying from 0 to pi.
- It is a double integration.
- Once we integrate R naught square into with respect to limits from 0 to R it is going to be 2 R cube by 3.
- At the same time integral of sin theta will be cos v.
- Then simplify this again.
- This is the y coordinate of the centroid, and here the centroid will be on the y axis.

Example or application from the lecture

- For this, I am going to consider a semicircle which is symmetrical about y-axis.
- If you consider.
- For this we have to consider one element.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

Before starting with the numericals, let's know about few basic formulas which we need for solving the numericals in Centroid.

Key details

- For this case, in the first case, where will be the centroid?
- The centroid will be somewhere here.
- Then what is X and Y coordinates of the centroid?
- X bar will be this distance and y bar will be this distance.
- Since the point is in the first quadrant, both x bar and y bar will be positive.
- Look at this, there is a same rectangle of base V and the height, H.
- The centroid is here.
- Since the centroid is in the second quadrant, the x coordinate of the centroid will be negative, and the y coordinate of the centroid will be negative.
- Since the centroid is in the third quadrant, both x and y coordinates of the centroid will be negative.
- That's why it is minus b by two and y by is also minus h by two.
- This is how the values will change depending on the position of the centroid.
- Where will be the centroid here?
- From the derivation, we know centroid will be at a distance of H by 3 with respect to the base of the triangle.
- Then what about with respect to apex of the triangle?
- With respect to apex it will be 2 h by 3.
- And here, since the x-axis is passing through the base of the triangle, y coordinate of the centroid will be h by 3.
- Then what is X bar from the derivation we know X bar is this distance from ear to ear distance.
- That will be equals to base by 3.
- And what is Y bar?
- Y bar is this distance, y coordinate of the centroid.
- That will be h by 3 because x is passing through the base of the triangle.
- And one more, since point is in the first quadrant, both x bar and y bar will be positive.
- Where will be the centroid?
- Then what is X bar?
- X bar is the X coordinate of this C.
- This is x bar.
- Since the point is in the third quadrant, x bar will be negative.
- Then what will be that value?
- With respect to this point, it will be 2b by 3.
- Whereas with respect to this it is B by three.
- Since y axis is passing to this point it will be two b by three minus two b by three.
- Then y bar will be.
- Since the x axis is passing through the ax of the triangle, then the y coordinate of central will be two h by three.
- Since the point is in the third quadrant, y bar is minus two h by three.
- Let B be the base and H be the height.
- Where will be the central?
- Then what is X bar?
- X bar is this distance from here to here.

- What will be that distance?
- 2b by 3 because from this side it is b by 3, from this side it is 2b by 3.
- And at the same time, since the centroid is in the second quadrant, x bar will be negative.
- Next, what is y bar?
- Why by the y coordinate of the centroid?
- That will be with respect to base, since x axis is coinciding with the base, it will be h by three.
- And the point is in the second quadrant, so y coordinate will be positive.
- Next is circle.
- For circle, we know the centroid will be at the center of the circle.
- Then what is X bar?
- X bar is this distance that will be equal to.
- R then what is y bar?
- That is y coordinate of this point C that will be also equal to R.
- And remember point is in first quadrant, so both are positive.

Example or application from the lecture

- The first one is let's consider there is a rectangle of base B and height H.
- Next, let's consider a rectangle.
- Next, let us consider there is a triangle of base B and height H.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

Of radius R, and where is the center of the circle?

Key details

- That is coinciding with the origin here, both X and Y.
- Then where will be the center here itself?
- If you look at the circle, the circle is symmetrical about this x-axis and this is the center of the circle.
- This is on the x-axis.
- Then what is x bar?
- X bar is this distance that will be equal to radius.
- Since the point is.
- On the negative side of the x axis, x bar will be minus r.
- Then what is y bar?
- Since the point is on the x axis, y bar is equal to 0.
- center here itself, then what is x bar?
- Since the point is on the y axis, x bar will be 0.
- Then what is y bar?
- Y bar is this distance.
- Since it is on the negative side of the y-axis, y bar will be minus.

- And what is this distance?
- Let O be the center of the semicircle.
- Whenever we come across semicircle, remember always we have to start with the axis of symmetry.
- Where will be the axis of symmetry?
- This is the axis of symmetry.
- Then where will be the centroid?
- On the axis of symmetry, somewhere here like this.
- X coordinate of the centroid.
- Since the axis of symmetry and y axis are parallel to each other, x bar will be this distance, that is r.
- And what is y bar?
- We know the distance of the centroid with respect to centre and the axis of symmetry is 4 r by 3 pi.
- Then look at this, the origin as well as the center of the semicircle are coinciding with each other.
- Centroid will be on the axis of symmetry.
- The OC distance is 4R by 3 pi, that we know that is X bar.
- 4 R by 3 pi.
- Since the centroid is on the x axis, y bar is equal to 0.
- Whenever you come across a semicircle, first thing is you have to locate the center of the semicircle.
- This O is the center of the semicircle.
- Then draw the axis of symmetry, then the centroid will be on the axis of symmetry.
- Then the OC distance is 4R by 3 pi, that is from the derivation.
- But what is X bar?
- X bar is this distance from the origin till C.
- Then what is this?
- This will be equal to total that is radius minus O C distance that is 4R by 3 pi.
- And since the point is in the second quadrant, x bar will be negative.
- Next, y bar.
- Y bar, since the point is on the x-axis, y bar will be zero.
- Now, first locate the center of the semicircle.
- This is the center.
- Then draw the axis of symmetry.
- This is the axis of symmetry.
- Next, where will be the center?
- We know from the derivation, this distance OC is 4r by 3 pi.
- But what is X bar?
- This is X bar that is equals to 4R radius.
- Y bar is this distance that will be equals to total R minus 4R by 3 pi.
- Since the point is in the.
- Fourth quadrant y coordinate will be negative.
- Next one is sector.

- We know in this figure.
- The sector is symmetrical about y-axis.
- So x-coordinate of this centroid will be 0 because the centroid is on the y-axis itself.
- And what is y-coordinate of the centroid?
- We know from the derivation of centroid, the distance of the centroid with respect to center on the axis of symmetries.

Example or application from the lecture

- Next consider a circle like this.
- Let's know about the semicircle.
- Let us consider there is a semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 20:00–24:59

Two by third of the R sin alpha by alpha.

Key details

- That's why y bar is this.
- In this case, where will be the centroid?
- On the axis of symmetry.
- Then what is X bar?
- This distance is x bar, then what is that?
- That is equal to 2 r sine alpha by 3 alpha.
- Then what is y bar?
- Since the point is on the x axis, y bar is equal to 0.
- That's it for today's class.

Example or application from the lecture

- Next, consider a sector which is symmetrical about x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

4 CLASS 36 Centroid_ Numerical

Study focus: CLASS 36 Centroid_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In today's class let's solve few problems in centroid.

Key details

- This is the first question in today's class.
- Problem 5/47.
- Determine the coordinates of the centroid of the trapezoidal area as shown in the figure.
- We have been asked to locate the centroid of this lamina.
- So what we can do here is we can use the method of approximation and we can locate the centroid of this lamina.
- To use the method of approximation, we know the formula x bar is equals to summation of EI Xi by.
- Total area y by summation of a i yi divided by total area.
- To use that formula, we have to divide this into elemental areas.
- How to divide this?
- Just draw a vertical line passing through this point.
- We are going to get one rectangle and one triangle.
- Together we will get a trapezium.
- The first one is rectangle.
- And what will.
- The width of the rectangle is 14 and the depth of the rectangle is 60.
- Then the second elemental area is.
- triangle here base of the triangle will be total 60 minus 40.
- After that the height of the triangle will be the height of the rectangle that is 60 ml.
- Next, we know the formulas x bar and y bar.
- Remember, since we have your two elemental areas, the formula is going to be a1 x1 plus a2 x2 divided by total area that is a1 plus a2.
- it by total area.
- That's what we have to find it out to find the value of x bar and y bar.
- A1 is nothing but area of the first elemental area.
- For time being, forget about this triangle.
- That will be equal to B into H.
- What is B from here?
- What is H?
- Here you will get it as 2400 mm square.
- Next, we need X1 and Y1.
- What is X1 and Y1?
- If you look at this rectangle, again it is a lamina.
- Each and every lamina will have a centroid.
- That will be this centroid.
- C1 stands for the centroid of this rectangular lamina.
- The coordinates of the C1 are X1, Y1.

- X1 is X coordinate of C1, Y1 is Y coordinate of the C1.
- Then what is this x1?
- We know for a rectangle from here the centroidal distance will be b by 2.
- Here b is 40.
- What is y1?
- From ear to ear distance that will be total will be sixty, sixty by two will be y one, that is h by two.
- You will get it as 30 mm.
- So we are done with the first elemental area.
- Next come to the second one.
- A2 will be the area of this triangle.
- What is A2?
- Half into base into height.
- Base is 20 and the height is 60.
- Substitute here, you will get the value of triangle A2.
- After that, we should know the value of X2 and Y2.
- X2 and Y2 are nothing but the coordinates of the centroid of the second lamina.
- Since this is a triangle here it will be having a centroid that is C2.
- The x coordinate of C2 is X2, y coordinate of C2 is Y2.
- Then how to find X2?
- Remember the triangular formula.
- From this point to here, it will be B by 3.
- But what is X2?
- X2 is from the y axis.
- That is X2.
- What is base here?
- Base is nothing but base of the triangle, that is 20.
- Substitute here, and we will get the value of X2 as 46.67.
- Next, we should know the value of Y2.
- We know the standard formula with respect to base of the triangle, the y coordinate of the centroid will be h by 3.
- H is 60.
- The formula is going to be like this: substitute the values of a1, x1, a2, x2 in this, and you'll get the value of x bar.

Example or application from the lecture

- This is the given lamina.
- Let's know about the dimensions of this rectangle.
- Now let's find out what is a1, what is x1, what is y1, next a2, x2, y2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Substitute the values of a1, y one, a2, y2, you'll get the value of y bar.

Key details

- You'll get test twenty-eight point.
- The coordinates a centroid.
- The coordinates of the centroid of the trapezoidal area are 25.33 comma 28.
- That is what the meaning.
- Next question 5 slash 47.
- Determine X and Y coordinates of the center of the shaded area.
- We have been asked to locate the centroid of this lamina.
- And look at this, the y-axis is here at a distance of 50 with respect to this member of a rectangle.
- Again, we don't have a standard formula for this figure.
- We have to use the method of approximation.
- To use the method of approximation, we have to divide this into elemental areas.
- That's up to you.
- Here I have taken it as three elemental areas.
- One is this rectangle, one more is this rectangle, and one more is a triangle.
- The first one is a rectangle, here width of the rectangle is 50, and the depth of the rectangle is 200.
- Second one is again a rectangle of width 150 and the height or a depth 200.
- Next third one is a triangle of base 100 and the height 200.
- Now we know the formula in method of approximation.
- X bar is summation of AI Xi by total area, y bar is summation of AI yi divided by total area.
- Since here we have three elemental areas, the formula is going to be a one x one plus a two x two plus a three x three by total area that is a one plus a two plus a three.
- A1 Y1 plus A2 Y2 plus A3 Y3 divided by total area that is A1 plus A2 plus A3.
- First A1 is nothing but AF the first.
- Element since it is in the form of rectangle, area of this rectangle will be B into H.
- Next, what is X1 and Y1?
- This lamina will have a centroid.
- For time being, forget about this second rectangle and this triangle.
- This rectangle is the area so it will have a centroid C here.
- Some are here C one we will call it as.
- Then the X and Y coordinates of this centroid are nothing but X1, Y1.
- What is X1?
- The distance of this C1 with respect to y axis and y1 is the distance of the c1 with respect to x axis.
- Where is the c1?
- It is in the second quadrant.
- Will be positive.
- That's the reason.
- And x will be equals to b by 2.

- We know the value of b 50 from here, substitute here, 50 by 2 will be equals to 25.
- Since it is in second quadrant, x coordinate will be negative.
- And y1 will be h by 2.
- That will be hundred.
- Then come to the second one.
- Area of this will be equals to B into H.
- Here B is one fifty and H is two hundred.
- Substitute that you will get the value of A2.
- Okay, now for time being forget about first elemental area and the third one, this triangle.

Example or application from the lecture

- This is the given lamina area.
- Let's divide this into elemental areas as rectangle and triangle.
- Yeah, what you can do here is either you can go for only one rectangle of base 200 and you can consider this is one rectangle of 50 and one more rectangle of 150 and one triangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Just consider about this rectangle.

Key details

- This rectangle is again an elemental area, so it will have a centroid that is C2.
- X2, Y2 are the coordinates of this C2.
- Then what is X2?
- X2 will be B by 2.
- B we know 150.
- And Y2, we know that is Y2 is equal to H by 2.
- Here H is 200.
- We are done with the first two areas.
- A3 stands for area of the triangle that will be half into base into height.
- Base is 100 and the height is 200.
- Substitute you will get the value of A3.
- Again, it's a lamina, it will have a centroid that will be C3.
- The coordinates of this C3 will be X3 from year to ear distance and Y3 will be this distance.
- Then how to find this X3 year?
- We know from the standard formulas the distance.
- Central with respect to this line will be B by 3.
- Please remember the formulas.
- But X3 is measured with respect to y-axis.
- From year to year it is 150 plus b by 3.
- And y three we know for a triangle the y the distance of the centroid with respect to its base will be h by three.

- Plus into minus you will get this entire term as negative, whereas the second term is positive, third term is also positive.
- Just simplify this, you will get the value of x1.
- Then y1 is.
- If you simplify you'll get the value of y1 as 93.34 mm.
- The coordinates of this android are 76.67 millimeter and 93.34 millimeter.
- That's the point where we assume the entire area of the lamina is concentrated.
- This is the next question 5 slash 51.
- Determine X and Y coordinates of the centroid of the shaded area.
- What is shaded here?
- It's not like a trapezium.
- In this trapezium, the circle has been removed.
- Then how to locate the center end of only this remaining area.
- Again, we don't have a formula, standard formula, so we have to follow the method of approximation.
- To follow the method of approximation, we have to divide this into elemental areas.
- Think how to get this shaded area.
- Earlier in one of the previous problem we just divide the trapezium into rectangle and triangle.
- But here if we divide this as triangle and rectangle, will we get this shaded area?
- Then in that what do we need to detect?
- We need to detect this circle.
- I am going to draw a horizontal line passing through this point for Tm.
- Then only we will get this shaded area.
- Will it be fifteen?
- No, because here it is 40.
- This is 40, these two are paralleline, so this height will be 40.
- Then what will be the height of the triangle?
- Total minus 40.
- From ear to ear it is 80.
- 80 minus 40 will.
- Give us 40 mm.
- Don't take it as 50.
- Here the first one is a rectangle of width 100 mm and height 40 mm.
- Second one is this triangle, base of the triangle will be 100 and the height of the triangle will be 40.

Example or application from the lecture

- Now let's start with method of approximation.
- Now let's start with the first one.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

Third one is deduction, remember that.

Key details

- And it's the circle of radius.
- The formula remains the same summation of AI xi by total area, summation of AI divided by total area.
- A1 stands for area of the first elemental.
- Since it's the rectangle, area will be b into h.
- Next, we need to find the value of x1 y1.
- Since it's the lamina, it will have a centroid that will be C1.
- C1 stands for centroid of this rectangle.
- Then x1, y1 are the x and y coordinates of this c1.
- Then what is x1?
- It will be since it's a rectangle, we have the y axis here, so it will be equal to b by 2 and y1 will be equal to h by 2.
- Next, we have to start with the second area.
- Here the second one is a triangle.
- Next, we know the value of base and we know the height of the triangle.
- Substitute here you will get the value of area of the triangle.
- When we talk about.
- The triangle we should forget about other areas.
- This triangle is again a laminary area, it will have a centroid that is C2.
- X2, Y2 are nothing but X and Y coordinates of this C2.
- Then what is this X2?
- We know in a for triangle the formula this x coordinate with respect to this side will be B by 3.
- Then what is y2?
- Y2 is this distance total.
- But in the standard formula, we know this distance, that is the distance of the centroid with respect to its basis H by 3.
- That is from this to this is h by 3.
- But what is y2 total.
- Then the last one is circle.
- Since area we have to deduct here, area doesn't exist here, it should be taken as negative.
- As being removed from this trapezium, so area itself is negative.
- area is minus pi r square, you will get less 1256.63 mm square.
- It will have a centroid that will be C3, then the coordinates of that will be X3, Y3.
- The sign of X and Y coordinates of the centroid depends upon where is the centroid, in which quadrant the centroid is.
- Since the centroid is in first quadrant, both X3 and Y3 will be positive.
- Then what is Y3 here?
- Just substitute the formulas.
- The formula is going to be a1x1 plus a2x2 plus a3x3 by a1 plus a2 plus a3.

- That is the general formula.
- I don't think you will get any doubts in first two terms.
- Once you simplify you take the value of x bar as 45.62 mm.
- substitute all the values from this tablet column and then.
- You'll get the value of y bar.
- Y bar is thirty one point three seven.
- That's the point where we assume the entire area is concentrated.
- Expir by bar.
- That's about today's class.
- If you come across any doubt, you can post your queries in Pesum forum.

Example or application from the lecture

- Now let's start with the method of approximation.
- Let's start with the first elemental area.
- Just consider this rectangle.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

5 CLASS 35 Centroid_ Numerical

Study focus: CLASS 35 Centroid_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

How to locate the centroid of the given figure.

Key details

- This is the question 5/55.
- Locate the y coordinate of the centroid of the shaded area.
- Look at this figure.
- In the sector, they have removed this triangle, and we have been asked to locate the centroid of only this shaded area, and that to only the y-coordinate.
- They have not asked about the x-coordinate.
- If they ask about x coordinate, then where will be the centroid?
- Look at the figure.
- Due to symmetry, the centroid will be on the y axis so x coordinate of the centroid will be 0.
- Since they have not asked, it's not necessary to mention about x bar.
- We need to find only the value of y bar.
- Next, how to get this shaded area?

- To get the shaded area, we have to use the method of approximation.
- We have to divide this into elemental areas.
- We can divide this as a sector and remove this triangle from the, then you will get this shaded area.
- There is a sector, the inscribed angle will be 90 degrees, nothing but pi by 2.
- Then radius is A.
- That's about sector.
- Then come to this triangle.
- Here the height of the triangle will be H.
- Then what will be this base?
- What will be this base of the triangle that we don't know?
- I am going to call the basis B since B is the term which I introduced in the.
- Question, I should know the value of B in the form of H.
- Then how to express B in the form of H.
- If this is 45 degrees, this angle will also be 45.
- Then due to symmetry, if total is B then this will be B by 2.
- Tan 45 is equals to b by 2 by H.
- Sorry, we will get the value of B.
- B is equals to 2 into H.
- First one, let it be a sector of radius A, and the angle alpha will be here will be equal to this will be alpha.
- In the sector derivation total angle was 2 alpha.
- Here also 2 alpha is pi by 2, that is 90 degree.
- Next triangle, second one.
- Base of the triangle will be 2 into h and the height of the triangle is h itself.
- And we know the formula for y bar, summation of a i yi divided by total area.
- area of the first elemental element.
- What is the formula to find area of a sector?
- R square into alpha.
- Remember alpha should be in terms of radians.
- This A1 area of this sector.
- Then we should do the value of Y1.
- What is the meaning of Y1?
- This sector will have a centroid that is C1, then the Y coordinate of that C1 will be Y1.
- Since the sector is symmetrical about y-axis, the centroid of the sector will also be on the y-axis.
- Then this is y1.
- How to find this y1?
- To find the Y1, we know the distance of the centroid of this sector with respect to its center and the axis of symmetry is 2 R sin alpha by 3 alpha.
- This is the formula standard formula.
- Next is triangle.
- Since this triangle is removed or it's not shaded, area should be taken as negative.
- What is base.

- Two times the dark h and the height is h itself.
- Substitute here, you will get the value of A2S minus h square.
- After that, we should know y2.
- What is y2?

Example or application from the lecture

- Here, they have given a sector.
- Then, before getting into the method of approximation, let's know about the dimensions of the sector as well as the triangle.
- For that I am going to consider only this part of the triangle, that is right angle triangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

This triangle will have a centroid that will be C2.

Key details

- The y coordinate of C2 is Y2.
- Look at this axis, x-axis is passing through the ax of the triangle.
- That will be 2H by 3.
- The general formula is a1 y1 plus a2 y2 by a1 plus a2.
- But since the value of A2 is negative, substitute that as it is here.
- That's why you will get the second term as negative here as well as in the denominator also.
- Once you simplify you will get the value of y bar.
- This is the value of y bar.
- That's the point where the entire area of the shaded region is assumed to be concentrated.
- Determine the coordinates of the centroid of the shaded area.
- This is the shaded area.
- We have been asked to locate the centrate of only shaded area.
- Then how to locate the centr of only shaded area?
- Here also we have to use the method of approximation, divide this into elemental areas, then use the formula of x bar and y bar, find out x bar and y bar.
- How to divide this into elemental areas.
- First if you draw a vertical line passing through this point, this will be triangle, then.
- This is semicircle, so the remaining will be rectangle here, then semicircle.
- In that we have to deduct this rectangle, then semicircle of radius 30.
- Only then we will get the shaded area.
- So this is the first one triangle.
- Second one is this rectangle.
- And the third one is this bigger semicircle of radius 40.
- In that we have to deduct fourth one, that is semicircle of radius 30.
- Fifth one is also deduction, that is a rectangle of width 50 and the height.
- Then only we will get this shaded area.

- We know the formula x bar and y bar a1 y1 plus a2 y2, so on divided by total area.
- First one is the triangle of base 40 and the height.
- Height will be the this that will be diameter of this bigger semicircle A T.
- That is area of the triangle.
- Then we need to find X1 Y1.
- X1, Y1 are the coordinates of the centroid of this triangle.
- The centroid of the triangle will be somewhere here, that will be C1.
- The x coordinate is this, that is x1.
- Then how to find this x1?
- We know the standard formula, this x1 will be equals to with respect to this side, it will be two times the of base by three.
- Next, we should know the value of y1.
- With respect to base of the triangle, y coordinate will be h by 3, here h is this AT.
- We are done with the first elemental area.
- The width of the rectangle will be how much?
- Total will be 140 minus 40, because this 40 is included in the triangle.
- Next we should know X2 Y2.
- Ah, this rectangle will have a centrite somewhere here at the center, C2, x coordinate of this, that is the distance from y axis to c2, that is x2.
- This is x2.
- Then how to find this x2?
- From the derivation we know from ear to ear distance will be b by 2.
- But x2 is from y axis, so we have to add this 40.

Example or application from the lecture

- Let's move on to the next problem 5 slash 58.
- Let's start with the first one.
- Okay, let's start with the first one.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then y2 is the y coordinate of this c2, this distance, that will be equal to total 80, 80 by 2 will be 40.

Key details

- We are done with the second lamina.
- Come to the third elemental area.
- The third area is semicircle of radius forty.
- I R square by 2 R is 40.
- You will get the value of area of this bigger semicircle.
- Next, this bigger semicircle will have a central.
- Where will be that centroid?

- On the axis of symmetry, somewhere here.
- If you draw a horizontal line passing through the center, this will be the axis of symmetry.
- Then what is X3?
- X3 is this distance.
- From y axis to C3.
- Plus what is this distance?
- That will be 4 r by 3 pi.
- Since it's a we are talking about bigger semicircle, radius is 40.
- Next y 3.
- Y3 is y coordinate of this C3 that will be equals to the di radius itself that will be 40.
- Next is the deduction part, that is fourth one, we have to deduct this smaller semicircle of radius 30.
- Next, where will be the centroid of this smaller semicircle?
- Again, that will be on the axis of symmetry somewhere here, C4.
- Then what is X4?
- Ear to ear distances X4.
- Then how to find the 64?
- From year to year we know 140 plus center to centroid will be 4r by 3 pi.
- Since C4 is in the first quadrant, it will be positive.
- 140 plus 4 R by 3 pi ERR is 30, you will get the value of X4.
- Then what is Y4?
- Y4 will be the diameter of sorry radius of this circle that will be 40.
- Next, the last is the deduction of this rectangle.
- What is b here?
- 50 and what is the height of this rectangle?
- You'll get it as minus 2000.
- Then we should know x phi and y phi.
- X phi and y phi are the.
- Coordinates of the centroid of this C phi.
- Then where will be what is this x phi?
- This distance is x phi.
- Then how to find that?
- That is 40 from here to here 40 plus 20 plus what is this?
- 50 by 2.
- We will get it as 85.
- Then we should know y2, sorry, y phi, y phi will be equals to.
- This 80 by 240.
- We got the value of all the terms in the equation.
- In this formula, so substitute here a1 x1 then plus a2 x2 plus a three x three plus.
- Then you should open the bracket, since it is minus it should be.
- Minus 450 pi into plus of 152.7.
- Similar next plus of minus 2000 into 85 divided by total area will be a one plus a2 plus a three.
- Since it is minus it will be minus four fifty pi 450 into pi, then minus two thousand.

- Once you simplify it, get the value of x bar.
- A1 Y1 A2 Y2 A3 Y 3 A4 Y 4 A5 Y 5 You'll get the value of Y bar.
- The Y bar is 37.55.
- This is the point 88.67, 37.55 will give us the coordinate of the centroid of the shaded area.
- That's the point where we will assume the entire area is concentrated.
- That's about today's session.
- If you come across any doubts, you can post your queries in the PESO forum.

Example or application from the lecture

- Then what will be this total distance that will be equals to from year to year 140 is given here.
- try to solve the problems given in the assignment as well as in the cushion bank.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

6 CLASS 29 Introduction to Beams

Study focus: CLASS 29 Introduction to Beams. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

When you are in the basement of any multi-story buildings like schools, apartments, or even in Mars, you can see some part of the building will be projected below the roofing, just like this, like a horizontal member.

Key details

- You can see in this picture these horizontal members, those projections, what you can see in the basement, we call them as beams.
- These beams play very important role in all the structures.
- Then how to define these beams?
- We are going to define beam as a structural member which offer resistance to bending at the rotation due to the applied loads.
- And these beams are usually long prismatic bars and the loads are applied normal to the axis of the bar.
- The first one is roller support.
- This is the roller support here.
- You can see the beam is resting on this roller support.
- The reason is if you think of applying a.

- Vertical load here, we can't move the beam along y direction because the roller will restrict the movement of this beam along the y direction.
- Whereas if you think you are going to apply a horizontal force on this beam like this, definitely it is going to move along the horizontal direction.
- Because there won't be any restriction from the support onto the movement of the beam along X direction.
- If you think of applying a couple at this free end, it is free to rotate about this roller support.
- That's why.
- Remember whenever we have a roller support, roller will offer resistance for the motion only along y direction, that's why we will be having only one reaction normal to the surface of the contact.
- This is about roller support.
- Next one is fixed support.
- In fixed support, the beam will be like this.
- It is represented in this way.
- Beam along y direction.
- At the same time, if you think of applying a horizontal force on this beam, again the.
- Support will offer restriction for the motion of the beam along X direction.
- That's why it will be having two reactions.
- One will be along X direction, one more will be along Y direction.
- In the same way, if you think you are going to apply a couple here at the free end R due to the application of the force.
- Do you think the beam will start to rotate like this about the axis of the beam?
- No, it won't, because.
- This will have a resisting couple.
- Three reactions to our forces, one along X, one more along Y, and one more will be the resisting couple like this.
- Next one is spin arranged support.
- This support will offer resistance for the motion of the beam in both along x direction as well as along the y direction.
- That's why two reactions, one along x direction and one more along y direction.
- This slide is very important to solve the numericals.
- Next is different types of beams.
- We are going to broadly classify the beams as statically determinate beams and statically indeterminate beams.
- Statically determinate beams.
- We can find the support reactions using the conditions of equilibrium.
- And in one more way, if the beam is having just sufficient number of support reactions which are enough for the beam to be in equilibrium.
- That is also called as statically determinate beam.
- If the beam is having more.
- Number of support reactions that necessary for the stable equilibrium configuration that is called as statically indeterminate beam.

- The first one is simple beam.
- If one end of the beam is on a pin connection or a hinged connection, and if one end is an roller, that we call it as a simple beam.
- How many number of support reactions here?

Example or application from the lecture

- We have different before moving on to the different types of beams, let's know about different types of supports.
- Ah when we draw the free body diagram in the numericals, if there is a roller support, usually we consider there will be only one reaction from the roller support.
- Let's talk about statically determinate beams.

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then at B it's roller support, so one reaction that is RB reaction at B.

Key details

- We can find all the three reactions using three conditions of equilibrium.
- That's why the name statically determinate B.
- This is the first one simple beam.
- The second one is cantilever beam.
- If one end of the beam is fixed like this and the other end is free, it is not resting on any of the support.
- It is called as a cantilever beam.
- Since it is a fixed support, there'll be two forces, one along its direction,.
- One more in a Y direction and one more is the resting momentum.
- And if the answer is positive, our assumption is wrong.
- The last one is combination.
- This is also called as overhanging beam because this part of the beam is extended beyond the support, hence it is called as overhanging beam.
- It is pined connection.
- Then it's a roller, only one reaction along Y direction.
- These three are the statically determinate beams.
- Next, we have statically indeterminate beams.
- Remember, in this we can't find all the support reactions using the three conditions of equilibrium.
- The first one is continuous beam.
- If the beam is having more than three support reactions, three are more than three support reactions, it is called as continuous beam.
- It is a pinned connection, so two reactions, one along X and one more along Y, AX, AY.
- Next at B it's a roller support, so only one reaction.
- At C again it's a roller support, so one more reaction.

- Total we have four reactions Ax, AY, RC, and RB.
- Hence it is statically indeterminate B.
- The next one is fixed B.
- If both ends of the beam are fixed, it is called as fixed beam.
- See here it is fixed here and here also it is fixed support.
- Hence it is called as fixed beam.
- How many number of support reactions?
- Fixed support will offer.
- Three reactions, two forces and one more is moment, and here also two reactions and one more is moment.
- As I said before, the direction depends upon us how we are and we can assume the direction of the forces as well as the moment.
- Next one is end supported cantilever.
- This is also called as propped cantilever.
- Nothing but if one end of the beam is fixed, the other end is resting on a roller, it is called as end supported cantilever, a propped cantilever.
- Since it is a fixed support, three reactions and at B it's a roller support, so one reaction RB.
- Total we have four reactions, hence it is statically indeterminate B.
- This is about different types of beams.
- Next, the different types of loading on beams.
- The first one is point load.
- This is also called as concentrated load.
- As the name itself indicates, if the force acting on a beam is acting at a particular point.
- are the area over which the force is acting is very small compared to the dimension of the beam.
- We call that beam as a point load or concentrated load.
- Look at this figure there is a simple beam and.
- The load is acting at a particular point here, hence it is called as a point load or concentrated load.
- When we solve the numericals, we want all the loads to be point loads for our calculation purpose.

Example or application from the lecture

- Let's count the number of support reactions.
- You can consider it as either clockwise or counterclockwise, no need to worry about the direction.
- If you consider clockwise.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

And this point load can be either inclined, vertical, or horizontal.

Key details

- Few times it can be horizontal.
- A few times it can be inclined.
- Whenever they give the inclined load, remember we know how to resolve the load along.
- Force along X and Y.
- Same concept here also we are going to apply and find the components of that load.
- Nothing but force along X and Y direction.
- The second one is UDL.
- This is also called as rectangular loading, uniformly distributed loading.
- We are going to represent the loading like this.
- The blue color is the beam, and this is the load that is acting on this beam.
- The intensity, this is the assume this is the starting point of the load and this is the end point of the load.
- From ear to ear, the intensity is uniform throughout, hence the name uniformly distributed load or rectangular loading, and here W stands for the height of this rectangle.
- And L stands for the base, nothing but the span of the UDL.
- Then, as I said before, we want all the loads to be point load or concentrated loads.
- Then how to convert this UDL into a point load.
- That point load should produce the same effect as that of this UDL on a beam.
- There should not be any net change on the.
- Then how to get that?
- For that to find the resultant of this UDL, there is a formula.
- Since it is in the form of rectangle, resultant will be equal to W into L nothing but base into height of the rectangle.
- Once you find the resultant, this is the.
- Point load R is equals to W into L.
- Where exactly this point load will be acting because UDL is acting over a span of L meters.
- Then wave wheels are acting.
- Remember, since it's the rectangle from the concept of centroid, it will be exactly at the midpoint L by 2 from here.
- L by 2 from here.
- Here this R which is acting at a distance of L by 2 will produce the same effect as that of the UDL acting over a span of L meters.
- That's what the meaning.
- U VL that is uniformly varying load.
- This is also called as triangular load.
- Look at this figure, there is a beam.
- This is a beam and there is a UVL.
- The intensity of the loading is varying from 0 to W over a span of L meters.
- And this W will be either in kilonewton per meter or Newton per millimeter.
- Next, as I said before, we want all the loads to be point or concentrated loads.
- Then how to find the resultant of this load?
- Remember, since it is in the form of triangle, we are going to use the formula of.
- Triangle half into base into height.
- The resultant of this will be half into base into height, that is W into L by 2.

- You will get the magnitude of the resultant of this loading.
- Next, where exactly that will be acting?
- From this side, it will be 2 times that of L by 3, and from here L by 3.
- That you should take it.
- If you are measuring from this side, it will be L by 3 from here to L by 3.
- Fourth one is combined loading.
- It's also called as trapezoidal loading.
- It's a combination of UDL and UVL, hence it is called as combined loading.
- Look at the figure here.
- There is a beam and.
- Ah there is a loading.
- At one hand the height of the loading is W1, at the other end it is W2, and the span over which the load is acting is L.
- We know the formula for rectangular loading and triangular loading.
- Just assume we are going to draw a horizontal line like this.

Example or application from the lecture

- And intensity of this loading can be either given in the form of kilonewton per meter or newton per millimeter, and L can be either in millimeter or meter, and L is the span of the UDL.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

For the time being, forget about the triangle.

Key details

- How to find the resultant of the rectangle?
- Base into height.
- What is height here?
- This is W1, so this is the height.
- Then what is the base?
- L so R1 stands for R1.
- One is nothing but we are going to assume the first loading is a rectangle.
- R1 is equal to W1 into L.
- We got the magnitude of the resultant of the rectangle.
- Next thing is where exactly this R1 will be acting, we know from the concept of centroid from ear to ear, L by 2.
- For time being forget about the rectangle.
- What is base here?
- It is the same.
- Then what about the height of this triangle?
- W2 is not the height because W2 is the height from Ear to B.

- Total height is W2.
- Then how to find the height of the triangle that will be equal to total W2 minus this one.
- W1 that will be the height.
- Triangle R2 is equals to half into W2 minus W1 into L.
- Next we should know the point of application or the action of this.
- The last one is general loading.
- In general loading, the intensity of the loading will be varying.
- And in the syllabus we have the numericals on point load, UDL, UVL, and combined loading.
- That's it for today's class.
- Try to solve the questions from the question bank and go through the assignment questions.
- If you come across any queries, you can post it in the PESU forum.

Example or application from the lecture

- Just consider the rectangle.
- If you consider triangle, to find the resultant of the triangle, we know the formula half into base into height.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

7 CLASS 30 Beams_Numerical

Study focus: CLASS 30 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000086 · EV0000087

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is question 5/101.

Key details

- Question is determine the reactions at A and B for the beam subjected to the uniform load distribution.
- Look at this question.
- There is a beam AB.
- The name of the beam is simple beam because one end of the beam is having pin connection, other end is having roller, and we need to find the support reactions at A and B.
- Since it is a pin connection, there will be two reactions, one along x direction, one more along y direction.
- We need to find what is ax and what is a y.

- Next, at B, since it's a roller, there will be only one reaction that is reaction at B R B that we have to calculate.
- These three reactions will be because of this loading on the beam.
- What is the name of this loading?
- Remember it's UDL uniformly distributed loading that we call it as rectangular loading also.
- First thing is always we have to convert this UDL into a point load because in our free body diagram we want all the forces to be point load or concentrated loads.
- Then how to convert this UDL into a point load?
- What is the formula?
- Do you remember?
- The formula is the resultant of this UDL R will be equal to W into L.
- What is W intensity of this loading.
- And L is span over which it is acting.
- Here what is W?
- W is intensity that is 6 kilo Newton per meter and what is L?
- Span that is 300 meter millimeter.
- Just take care of the conversion.
- Whenever we want to draw the free body diagram, the first step is we need to project all these important points vertically downward just like this.
- This is the point where the load is ending.
- Just draw a vertical line, then draw a vertical line passing through B.
- Either start from left end of the beam or from the right end of the beam so that you will not miss any of the forces in the free body diagram.
- I am going to start from this end of the beam and I'm going towards right side.
- Look at the beam here.
- At EA it is having a pinned connection.
- Pinned connection means two reactions.
- This is the assumption we are going to make.
- Remember, if we get the answer as positive, our assumption is correct.
- If you are getting the answer as negative, then we have to change the direction of the arms.
- That's it.
- No need to worry.
- Next we have UDL here.
- The intensity of the point load is 1.8 kilonewton, so 1.8 kilonewton here.
- But where exactly this will be acting?
- Remember the formula for U UDL it will be at a distance of L by 2 from here and L by 2 from here.
- Since L is 3000 it will be equal to 150 mm and 150 mm.
- Next we don't have any loading from in between this three hundred millimeters, so no need to do anything here.
- Then come to B at B since it's the roller support.
- The roller will offer only one reaction that is.
- Once we are done with the free body diagram of a beam, we have to apply the conditions of equilibrium.

- Look at the forces here.
- They constitute general force system.
- There is only one horizontal reaction that is Ax.
- We don't have any other horizontal force or reaction.

Example or application from the lecture

- Let's get into the solution part.
- Since the load is given per meter and this is in millimeter, convert from meter to millimeter or millimeter to meter.
- Once we know the resultant of the loading let's start with the free body diagram of the bean A B.

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EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 05:00–09:59

So we are done with the first one.

Key details

- One is a y.
- This one and one more is RB this one.
- To avoid that I'll go I'll start with the third condition of equilibrium that is moment about any point zero.
- Remember this is sum of the moment of all the forces about A should be 0.
- Next, move in this direction towards right.
- After this, the next force is 1.8 kilonewton.
- Apply the right hand thumb rule.
- Once you apply the right hand thumb rule here, your finger thumb is going inward.
- If the thumb is going inward, that indicates it is.
- Clockwise and in one more way if you are moving in this direction to reach here, you have to take this turn.
- This is the shortest rule.
- Clockwise means negative.
- That's what is written here.
- We are done with the moment of 1.8 kilonewtons.
- From here to here we don't have any forces.
- Then come to RB.
- RB is acting in the counterclockwise direction.
- forces R B into perpendicular distance.
- What is the perpendicular distance from point B to A?
- It is 150 plus 150 plus 300.
- Once you simplify this, you are going to get the value of RBS positive plus 0.45 kN.
- What the plus sign indicates our assumption is correct.
- We assume the force RB is upward.

- That is what this plus sign indicates.
- We are done with the reactionate roller B.
- Next we are left with only one more unknown that is AY.
- Then how to find AY?
- We have one more condition of equilibrium that is sigma f of y is equal to 0.
- Apply this sigma f y is equals to 0.
- AY is upward.
- Then 1.8 is downward.
- RB is upward.
- That should be equal to 0.
- We know the value of RB that is 0.
- 45 do the substitution here.
- Then you simplify, you'll get the value of a1.
- You'll get the value of Ay as 1.35 kN and the answer is positive.
- That indicates our assumption is correct.
- Remember, since it is a pin connection, what we can do here is we can write the reaction at A as under root of ax square plus a y square.
- Ax is 0, so reaction at A will be equals to directly AY.
- That's it.
- This is the next question.
- 5 slash 103.
- Calculate the reactions at A and B for the beam loaded as shown.
- Do you remember the name of this beam?
- It is called as combination or overhanging beam.
- Combination is because it is having pin connection here and it's a roller.
- Overhanging is this part of the beam, this one meter is extended beyond the support B roller support B.
- Hence it is called as overhanging beam.
- What is the name of this loading?
- It is the combination of triangular and rectangular loading.
- Then how to divide this loading?
- We know if we draw a line ear, horizontal line passing through this, it will be a rectangle down and a triangle ear.

Example or application from the lecture

- Next, if you consider second condition sigma f y 0, we have two unknowns along y direction.
- If you consider Ax and AY, these two are acting at the point A itself.
- Consider it as plus.

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EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 10:00–14:59

Let's see how we can draw the free body diagram for this.

Key details

- First thing is we have to divide this into rectangle and triangle.
- To divide that into rectangle and triangle, just draw a horizontal line passing through this.
- For the time being, forget about this triangle.
- Is it nothing but rectangular loading are UDL uniformly distributed loading?
- Then how to find the resultant of UDL?
- For that we know the formula W into L.
- W stands for intensity of the loading and L stands for the span over which the load is acting.
- Here W is right now 400 because that is the height of this rectangle.
- And the span L is nothing but the distance over which it is acting.
- 6 plus 1 plus 1 that is 8 meters.
- Since this is also in meter and the distance is also in meter, no need to worry about the conversion part.
- So total the resentment of this first loading is 3.2 kilonewtons.
- The second one is triangle.
- Just forget about the rectangle.
- We know the formula R is equals to half into base into height.
- Base is here A 6 meters because the triangle is having a base steel here, that's it.
- What is the height of this triangle?
- Total is eight hundred.
- Then how to find this height of a triangle?
- Total eight hundred minus height of the rectangle.
- You will get this height.
- This is W L by 2.
- And one more thing.
- When you talk about R1, that is the resultant of this rectangular loading, where exactly will that be active?
- L by 2 distance with respect to a and L by 2 with respect to this end.
- Nothing but 8 by 2 meters.
- Then where will be the resultant of this triangle with respect to this side we know L by 3.
- If you are talking from this side, it will be 2L by 3.
- That's why I have written it as next.
- L by 3 with respect to A.
- 6 by 3 is equal to 2 meters.
- Next, we can now we can start with the free body diagram of the beam.
- Need to draw the free body diagram.
- Remember we need to project all these points vertically downward.
- This will be the beam AB.
- Start from any one of the end, either from this end, or from this end.
- What is the name of this connection?
- It is a pin connection.
- This is up to you the direction.
- We are done with this.
- Next what do we have?

- We have R2 from here 1.2 kilonewton at a distance of 2 meters.
- We are done with the resultant of the second one.
- Where is the resultant of the first one with respect to 4 meters.
- 4 meters means after R2 we are going to get that.
- After this, we are done with the resultant of the loadings.
- Next move in this direction.
- We don't have any other loading, but here at a distance of 7 meters with respect to A, we have a roller support.
- We are done with the free body diagram of the beam.
- Once we are done with the free body diagram of the beam, next thing is we need to apply the conditions of equilibrium.
- What is the first condition of equilibrium?
- Sigma F is equal to 0.
- Since there is no horizontal force acting on the beam, Ax will be straight away.
- it will be equal to 0.
- I'll start with the moment equation because if you apply sigma f y 0, we have two unknowns a y then r b.
- Just to avoid that I am going to apply moment about E as 0.

Example or application from the lecture

- Just consider this rectangle.
- For when you consider a triangle, you no need to consider the rectangle.
- Consider that as it is one meter.

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EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 15:00–19:59

Once you apply moment about E as 0, look at this at A we have two forces Ax and AY.

Key details

- Since these two are acting at the same point, moment of these two forces will be zero.
- That's the reason why I'm not considering Ax and AY.
- More in this direction, which is the next force we have?
- 1.2 kilonewton downward.
- Don't look whether it is downward or upward.
- When we are talking about moment you should think whether it is.
- Making clockwise rotation or anti-clockwise rotation with respect to point under consideration.
- If your thumb is going inward, so it is clockwise.
- Since it is clockwise it should be written as minus into what is force here?
- What is the perpendicular distance?
- Next comes the second force that is 3.2 kilonewton.
- First think about the direction clockwise or counterclockwise.

- It is clockwise with respect to A, so minus forces three point two into distances four meter.
- That's what minus 3 point two into 4.
- We are done with the moment of this force.
- Next, from ear to ear, we don't have any other force.
- Lastly, we have RB.
- RB, if you apply right hand thumb rule, the thumb is coming outward.
- Counterclockwise plus what is force?
- Then what is distance from B to A?
- What is the distance?
- Counted three plus four seven.
- Plus Rb into 7 is equal to 0.
- Once you simplify this, you will get the value of RB.
- To find a y we are left with one more equation sigma f y is equal to 0.
- Again start either from descent or from descent.
- Then minus 1.2 because it is downward.
- 3.2 is also downward, so minus 3.2 RB is upward, so plus RB should be equal to 0.
- Since we know the value of RB, that is 2.17, substitute that here, find the value of Ay.
- AY you will get it as 2.23 kilonewton.
- The answer is positive.
- Our assumption is correct, so a y is acting in the upward direction.
- Once you are done with this, remember reaction at A will be always equal to under root of a x squared plus a y squared.
- Since ax is 0, reaction at a will be a f itself and that will be on a y direction.
- You can go through the assignment questions, try to solve the assignment questions, and go through the question pack.
- If you come across any doubts, you can post your questions and queries in PESU forum.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

8 CLASS 31 Beams_Numerical

Study focus: CLASS 31 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question 5/105.

Key details

- Find the reactions at A due to the uniform loading and the applied couple.

- Look at this beam.
- The name of the beam is.
- Cantilever beam because one end is fixed and the other end is free.
- Hence it is called as cantilever beam.
- The name of the support here is fixed support.
- It will have three reactions, two forces, one along X, one more along Y, and one more will be the resisting moment.
- That can be either taken as clockwise or it can be taken as clockwise or.
- They have asked us to find the reaction at A, nothing but Ax, A by then resting moment at A we have to calculate.
- if you look at this, this is UDL, uniformly distributed loading.
- When we talk about the free body diagram, we need to convert this UDL into a point load.
- How to do that?
- We know the formula W into L.
- W will be equal to 2 kN and L will be.
- Where exactly that will be acting?
- Midpoint is nothing but three by.
- Two, that will be one point five meters.
- And this is moment of a couple it's a free vector.
- For discussion you need to start with the feedbody diagram.
- Before that, there is a.
- We know the formula to find the resultant R will be equals to W into L.
- What is W density of the loading that is 2 and L is span over which it is acting three meters.
- Where exactly the six will be acting?
- Since it's a UBN, it will be acting at a limit point.
- and only one point five meter from here.
- Here it's a fixed connection.
- Two reactions one three reactions to our forces A then AY There will be one more resting moment M A.
- Next, six we are done with this, then there's no loading and here twelve kilonewton meter.
- That moment is acting.
- Start with the conditions of equilibrium.
- So the first one is sigma above.
- is equals to zero.
- If you have this sigma if x is equal to zero, there is only one hard certain four.
- That is e x.
- Next apply the second condition sigma f of.
- Y is equals to zero.
- Since a is upward, so plus a y, then six is downward minus six.
- This is equal to zero.
- Then a y is equal to six kilon.
- The third condition is sum of the moment of all the forces as well as the couple above the point is zero taking counterquest positive.

- Then moment of these two forces is zero, but MA is nothing but the resisting moment because of the support.
- That is different sum of the [ASR low confidence; check audio].
- Since I have considered here counterclockwise, so plus.
- Yum vector don't multiply the distance.
- Next is six kilonute, update the right hand thumb rule.
- It is thumb is going inward clockwise.
- One point five.
- Then next if you go like this, lastly we have twelve kilometer meter.
- Since it is in a counterclockwise direction, it should be taken as plus twelve.
- It's already forced into distance, so don't multiply the distance.
- It's really vector.
- This should be bigger than zero.
- Three kilonewton meter.
- What this minus sign indicates?
- We assume the existing moment here at A is in the counterclockwise direction, but actually that is in the clockwise direction.
- Three kilonewton meter clockwise.

Example or application from the lecture

- First, let's start with the beam.
- Before that let's discuss about the type of loading.
- When we consider sigma f and sigma f y we should not consider this because this is moment of a couple.

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EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

These are the reactions at this about E.

Key details

- This is the next question 5/116.
- For the beam and the loading shown, determine the magnitude of force F for which the vertical reaction at A and B are equal.
- With this value of F, compute the magnitude of the pin reaction at A.
- Nothing but Ax will be equal to R B or one more way R B will be equal to Ax.
- Ah F is also a unknown.
- After that there's one more unknown that is Ax.
- And in this particular question, Ax will not be equal to 0 because this is the entire beam.
- Horizontal and vertical.
- This is a single beam.
- R B as well as F will be unknowns.
- So four unknowns means statically indeterminate.

- Yo what is vertical reaction at D is equals to reaction at the end.
- For what value of this F, if A is equal to R B we have to calculate.
- Just project all these points vertically downward and start with the B.
- we have this is E and this is B.
- Only the points load in the free body line, so I'm going to call it as one.
- If this is what I need to find the result end of that loading, I'll call it as R1.
- R1 is equal to since it is in the form of triangle, it will be half into base into height.
- W by two.
- Half into what is W intensity of 2 into length.
- Span over exactly UEL acting is 2.
- Next way exactly it will be acting from here to L by three from here.
- R1 that is equal to 2 kilonewt at a distance of from here 2 by 3.
- From here 2 times of 2 by 3.
- Next, here we have two repezoidal loading.
- One wall span of four meters and one more one more wall span of three meters.
- Then how to divide?
- Just draw a horizontal line here.
- R2 stands for the resultant point load of the rectangle.
- Formula will be W into L.
- What is W intensity?
- It will be eight kilonewton.
- Where exactly this way?
- 8 kilonewton will be at the midpoint.
- Since it is four meter, 4 by 2 will be equal to 2 meters, that will be less than the point A.
- Span will be 2 meters.
- This is two meters from here to here.
- Next, R three stands for the result of the triangle.
- Half into base.
- That so that will be four span into height will be equal to five.
- Total is five.
- That you'll get us three.
- Where exactly this tool twelve kilonewton will be here?
- From here two times the deaf L by three from here.
- L by three.
- How much will get?
- That will be left side of the point.
- T by point three.
- Two times the tough.
- From this side, two times the duff.
- Four by three.
- This is the distance.
- Three you will get it.
- Then from here eight by three.
- From here eight by three means to the calculation, you'll get it as two point double six seven.

- Two point double six seven means in between eight and.

Example or application from the lecture

- If you don't consider the given condition a y is equal to r b, this will become statically indeterminate because ax a y r.
- Remember we have to consider if I is equals to RB.
- Let's get into the solution part.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Point E we'll get this slowly.

Key details

- This is twelve kilometer.
- Remember it is between eight and eight.
- The distances from here to here it will be 2.67.
- Next, A we have a pin correction, two reactions, AY, then one more will be horizontal.
- This will be A.
- I have taken it left side because anyhow here it is that to balance that A should be towards left.
- One meter we don't have any loading, next comes the trap side loading.
- Again draw descent line passing through the this will be the fourth one and this will be the fifth one.
- Descendant of fourth one fourth will be because since it is a rectangle, W into L will be the descendant.
- What is W three.
- Because this is the height of the rectangle and L is three.
- Where exactly this 9 will be acting?
- That we have to calculate midpoint.
- Nothing but from this point, 1.5 meter.
- Then from here to here, one point five meter we have this R4 of nine kilo Newton.
- Next, after this, we need to find the resultant R5 will be equals to.
- Triangle so half into base into height base is three and the height will be total five minus three is the height of the triangle.
- That will be two.
- Where exactly this three will be acting from here L by three from here to L by three.
- L by three means four.
- L by three.
- L A ye is three.
- Three by three will be equal to one meter.
- Nothing but it will be left side of this nine kilonewtons.
- This is three kilonewt and the distance will be one meter.

- Next we have to add this three meters.
- We don't have any loading for the next two meters, take that as it is.
- Next year it's roller connection so only by the reaction RB.
- Yes, take it as it is.
- We are done with the free body diagram.
- Remember we have to write AY is equal to RB.
- Next we have to apply the conditions of equilibrium.
- Start with the first condition sigma f of x is equals to zero.
- 2 kilonewt is right side, so plus 2, and this is left side, AX.
- AX will be equals to 2 kilonewt.
- Our assumption is correct.
- We assume AX is towards left.
- That is correct.
- Next is moment equation.
- I will for moment about point is zero.
- Taking counterclockwise is positive.
- Start with this two kilonut.
- Thump is coming outward because the force is acting in this direction, so curl your fingers in this direction.
- It is counterclockwise, so plus two into perpendicular distance will be this is the line of action of the force.
- And this will be the perpendicular distance that is vertical by three.
- We are then the moment of this next is this A kilo Newtons.
- It is acting in the counterclockwise direction, so plus yet.
- And what will be the distance?
- The distance from here to here is two meters.
- Three minus two will use one meter distance.
- Next is twelve kilon, that is counterclockwise.
- Plus twelve distances from year to year it is three.
- T minus two point double six seven.
- Three kilonewton is acting in a clockwise direction.
- Next nine kilon counterclockwise.
- Minus nine so nine kilometer clockwise.
- Then comes RB.
- RB is in a counterclockwise direction.

Example or application from the lecture

- You are calculating moment about E, so consider this part of your palm RTA, and you have to apply the right hand thumb rule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

From here to here it will be two plus three five plus one six.

Key details

- Then F is in a clockwise direction.
- This will be 3 plus 25.
- 5 plus 3 8.
- 8 plus 1 9.
- This should be equal to 0.
- Simplify this, you'll get the value of you'll give one equation 6 RB minus 9 F is equal to 15.
- You'll get this expression condition of equilibrium that is sigma f of y is equal to zero.
- If you apply this.
- 8 is acting downward, so minus 8, then 12 is acting downward minus 1, 3 is acting downward minus 3, then 9 downward, minus 9, a y I left, so plus a y.
- Then plus RB.
- Minus F is equal to zero.
- Since E is equal to RB.
- Then you'll get two RB minus F is equal to 26.
- This is the equation two.
- Once you solve one and two, from one and two, you'll get the value of RPS.
- One eight point one kilo Newton.
- Then F as one zero three six.
- One zero point three six kilo Newton.
- This is the value of.
- AY that is one eighteen point one eight is with this I am going to conclude today's session.
- go through the assignment questions.
- If you come across any queries you can post your queries in PSU forum.

Example or application from the lecture

- Since it is horizontal, only to consider this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

9 CLASS 41.1 Beams_Numerical

Study focus: CLASS 41.1 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the first question for today's class.

Key details

- Determine the reactions at A and B for the beam loaded as shown.
- There is a beam AB.
- What's the name of this beam?
- Do you remember?
- This part of the beam, this 4 meter is extended behind the support, hence it is called as woven hanging beam.
- It is also called as combination because it's a combination of pin, support, and roller reaction.
- A beam sorry This load 18 kilon per meter it is a UDL spread it over a span of four meters.
- Couple 15 kilon meter and it's a free vector.
- Remember this it's a free vector, even though they have specified some point like two meter here with respect to N with respect to B.
- It doesn't matter where it acts because it's a free vector.
- The moment of a couple is a free vector.
- The magnitude is 15 kilon meter.
- Now the first is to find the support reactions AX AY then reaction at B.
- So the first thing is we have to start with the free body diagram.
- See, before that we have to convert this UDL into a point load in the free body diagram.
- How to convert UDL into a point load?
- We know the formula, the resultant of the UDLR will be equal to.
- W into L A F rectangle.
- What is W here eighteen kilonewton per meter and what is L span over which UDL is acting four meter.
- The various 72 kilo newton.
- Next, where exactly the 72 kilo newton will be acting?
- Midpoint, nothing but with respect to B, L by 2, and with respect to the free end, L by 2.
- Remember we have to project all these points vertically downward, then draw a horizontal line, this will be the beam AB.
- Next start from any one of the end, either from A or from this free end.
- I am going to start with the end A.
- Since it is a pin connection, two reactions, one along X direction, one more Y direction.
- Next moving this way.
- Look at this, you have to move in this direction.
- Then after two meter, the mentioned the moment of a couple fifteen kilonewton meter, take that as it is.
- Next, after two meter we don't have any loading here.
- Then at B it's a roller support.
- that will be having only one reaction normal to the surface of contact so RB R B will be vertically upward.
- That way exactly that will be acting.
- As I said before, midpoint, L by 2 then L by 2.
- Yes, we have considered each and every force acting on the beam.
- How to find AXA by R B?

- Since it is a general force system, we have to apply the three conditions of equilibrium, sigma fx, 0, f y 0, then momentum about any point, 0.
- That's it.
- Next y direction we have two react two unknowns A by then R B.
- Before that I'll apply the moment equation.
- Moment about AS 0.
- This is the second condition of equilibrium.
- Look at this A.
- At A we have two reactions, but moment of these two will be zero.
- Then moving the.
- There is a moment of a couple 15 kN meter.
- Look at the direction, is it clockwise or counterclockwise?
- It is clockwise, so it should be taken as minus 15.
- And do we need to multiply the distance?
- Since it's kilonewtan meter, already the distance is there, it's a free vector.
- Don't multiply the distance per moment of a couple.
- Minus 15, we are done with the moment of this.
- Then come to this RB, RB is in the counterclockwise direction.
- Apply the right hand top rule.
- R B is in anti clockwise direction.

Example or application from the lecture

- First, let's look into this beam.
- Let's start with a free body diagram of the beam.
- Next four meters we have a UDL we have to consider the result end of that UDL that is 72 kilonewton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Next, which is an X load 72 kilonton.

Key details

- It is acting in a clockwise direction like this.
- Clockwise with respect to A.
- What is the perpendicular distance from this point to A?
- It is 2 plus 2.
- 4 plus 2.
- There is only one unknown in this equation that is RP.
- If you simplify you'll get the value of R B as positive.
- 111.75 kilo Newton.
- This plus sign indicates that assumption is correct.
- We assume the RB is up, but force that's correct.

- Next we are left with only one more unknown that is AY.
- Apply the sigma FY 0.
- So next is the RB plus R B because it is upward then minus 72 is equal to 0.
- Since we know the value of RB, substitute RB, you will get the value of Ay.
- Please make sure so AY is actually acting in the downward direction.
- Determine the reactions at A, B, and D for the pair of beams connected by ideal pin at C and subjected to the concentrated and distributed loads.
- Look at this figure.
- There are two beams A, C, and C D.
- These two are connected by IDL pin at C.
- If you think of finding the support reactions A, C, Y, R B, then.
- R D by considering beam A and D, we have more number of support reactions, then it will become statically indeterminate because we can't find all the reactions using the conditions of equilibrium.
- Then what we can do here is.
- We can divide this into two games, A C then C D.
- Here two reactions at A, then one reaction at B.
- Today we'll have like five support reactions when we talk about being AC in usually.
- Next option is being C D.
- If you think of C D, so two reactions at C, then.
- Only one reaction at D.
- This is how you have to solve this particular question.
- First one is considered being C D.
- Before that we have to find the resultant of this U V L in your point.
- What is the formula to find the resultant of UVL?
- We know the formula R is equals to half into base into height.
- Half into base into height.
- What is height here?
- 5 kilon meter.
- Then what is the base here span over which the load is acting?
- Next, where exactly this 10 will be acting from this end, nothing but from C, it will be at a distance of L by 3, or from this end it will be at a distance of 2 L by 3.
- First, project all the points vertically downward, draw a horizontal line, this will be the beam C D.
- Then start from any one of the end.
- I am going to start from this end, C.
- At C it's a pin.
- Next we have move in this direction towards right, we have UVL.
- What is the results of UV Lenkil Newton?
- Where exactly is that it is acting?
- L by 3 with respect to C.
- Resultant is 10 kilometers.
- Then what is the remaining distance from year to year?
- 4 minus 1.

- 3T that will be equal to 2.67.
- Then after 1 meter we have a roller support.
- This is the complete free body diagram of being CD.
- We are done with the free body diagram.

Example or application from the lecture

- So AY is upward, so plus AY don't consider this because this is moment of a couple, it's not a force.
- When you think of AC, consider the reactions.
- Then when you consider this AC separately, you have to consider the reactions at P at C.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

Start with the first one sigma fx 0, there is only one horizontal reaction, CS, so CS will be 0.

Key details

- Next I'll go for moment equation because y direction we have two unknowns.
- Then go in this direction towards right.
- The next force we will get is 10 kilometer.
- How is the direction of 10 kilometer?
- Look at this.
- It is clockwise with respect to C.
- Minus forces 10 into distances 1 point three.
- The next forces R D it is look at this the direction.
- Counterclockwise so plus Rd into distances, total distance that is 1 plus 4, 5 meters.
- We have only one unknown that is Rd.
- Simply 5 this you get the value of R d s 2 point double six seven kilo Newton.
- Next, we are left with one more unknown C by.
- Once you apply that, plus C by because upward, downwards, or minus 10, upwards or plus R D will be equals to 0.
- Since we know the value of Rd, substitute here, you will get the value of C by S.
- 7.334 kilogram and our assumption is correct because the answer is positive.
- CY we are taking it as upward.
- The joint C is in equilibrium, so whatever the vertical reaction exerted in this beam C D.
- C D CY is upward.
- That should be balanced by the reaction from the beam AC.
- Then only upward and downward C will get cancelled, the joint C will be in a state of equilibrium.
- Before that we have to convert this UV L into a point, or again we know the formula R is equal to W into L by W L by 2.
- W here is we know from here it is 3 kilon 5 kilonewton.
- Then span is 2 plus 2 that is 4 by 2 you will get the value of 10 kilometer.

- Next way exactly this will be acting with respect to C, with respect to C L by 3, whereas with respect to this point 2 L by 3.
- After that at A we have two reactions, horizontal, vertical.
- Next 3 meters we don't have any loading.
- Then we have this.
- Next we have the resultant of this UV, it is at a distance of L by 3 with respect to C.
- If it is 1.33, then with respect to this point it is going to be 4 minus 1.33 kilonewt.
- Already we know the value of C y here, that is 7.834.
- Three announce ax a y then r b.
- Start with the first condition of equilibrium sigma fx zero.
- Then go in this direction.
- The first course is twenty five.
- Look at the direction up in the right hand thumb rule.
- It is in the clockwise direction.

Example or application from the lecture

- Consider moment about C0.
- If when you consider moment about C0, don't consider C S and C Y because they are acting at the same point, moment will be 0.
- But remember when we consider the beam AC here, from the previous thing we got C by S 1.7334 kN upward.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

Just go in this direction.

Key details

- Next course is R B.
- Look at the direction up here right hand rule.
- It is counterclockwise direction.
- Next we have 10 kN.
- Look at the direction.
- It is clockwise.
- That will be equals to from year to year 2.67 plus 3 plus 3.
- That is a perpendicular distance for 10.
- Then look at C y it is clockwise so minus Cy into distances from year to year 4 plus 3 plus 3.
- Is last condition sigma f y 0.
- twelve point nine six kilonewton.
- Remember reaction at A will be equal to the root of A X square plus AY square.
- Since Ax is zero reaction at A is A by the answers are tableture.
- With this, I am going to conclude today's session.
- Go to the assignment questions, then question bank questions.

- If you come across any doubts, you can post your queries in PESO forum.

Example or application from the lecture

- And no need to consider CS because it is Z or it is on the same line of action.
- If you consider sigma f y 0, it is upwards, so plus a y then minus 25, plus b minus 10, then minus c y will be equal to 0.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

10 UE25CV141A_Unit 3_Class 53_AVSummary

Study focus: UE25CV141A_Unit 3_Class 53_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr Sv Vingitish from Civil Engineering under the Department of Science and Humanities, PS University.

Key details

- This is class number fifty-three.
- We are going to be starting the new topic in unit three under distributed forces, that is Teams internal effects.
- Written by Miriam and Craig.
- It's the eighth edition and SI Mosh.
- Chapter five, Beams Internal Effect is the.
- Determinate and indeterminate based on the type of support and the number of supports that is going to be acting on the beam.
- If it is equal to the number of equations of equilibrium or less than the number of equations of equilibrium, it is determined B.
- If the number of supports, external supports, are more than the available number of equations of equilibrium.
- Then they are treated as indeterminate beams.
- However, the indeterminate beams can be determined or can be the support reactions can be determined using other equations.
- The external effect or the support reactions which we have found or which we know how to find, but now we are trying to find what are the internal forces that is going to be acting in the beam because of the application of the external forces and as well as the external support reactions.
- They are shear force, bending movement, and torsion.

- Simply called a shear bending untorsion.
- In addition to the.
- Tension and compression that the beam is going to take.
- A beam has to resist all these three forces that is shear bending and torsion.
- The first one indicates the shear force.
- There could be forces acting whatever is indicated here.
- This will make that section two separate out or it gets sheared off.
- If the forces are acting like this, then it is going to get sheared off or the beam is going to get separated.
- That is what is called as shear force.
- Next coming to bending moment, if we apply movement like this, okay.
- Then the beam is going to bend because of this bending, it is called as bending movement.
- Earlier we used to find moment because of all the forces that is acting on a particular beam.
- But the bending moment is we are going to find out the moment acting only to the left side of the section or to the right side of the section.
- Moment at any particular point is equal to zero as the one of the equations of equilibrium.
- That is going to be applied when we take entire beam as one, including the support reactions, the applied forces, everything put together, what is the moment that is acting at a particular point should be equal to zero.
- But whereas bending moment, we are taking only the left side of the portion and whatever is the load that is acting in the left side of the portion, because of that whatever is the moment that is going to be coming, that is treated as bending moment.
- Or instead of taking left side, we can also take to the right side.
- For a section, whatever is the force that is acting to the right side of the section on the beam, if we take the movement of all those forces at that section, then whatever value that we are going to get will be called as the bending moment.

Example or application from the lecture

- You can see the different diagram that is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 05:00–09:59

This beam is subjected to two movements.

Key details

- One is clockwise, the other one is anti-clockwise, at the two ends.
- Electrical wire between the two cables, something like this, it is going to bend.
- Just like your half moon.
- The top zone or the top fiber, the extreme top fiber would undergo a compression and it will be under reduction in the length and it will be under compression.
- Whereas the bottom one, the extreme bottom layer, if you take, it will go, it is going to elongate and that would be under tension.

- Is called as a neutral axis.
- The zone above the neutral axis or the neutral layer is under compression, and that is called as a compressive zone, whereas the zone which is below the neutral axis or the neutral layer is called as a tensile zone.
- Next, we have this that is the torsion.
- If the movement that is acting because of the application of the load along the axis of the beam, that is the longitudinal axis of the beam, that movement is referred to as torsion.
- The internal forces that is going to be acting are the combination of all these three, which is indicated here.
- There is bending moment M as well as there is torsion P torsional moment T acting along the longitudinal axis of the beam.
- Unlike the external effect where we used to have the supports, replace the support by its reaction, determine those reactions by using the equations of the curve.
- Here, when we have to find out the shear force of bending moment.
- We have to use a different sign convention.
- This is the sign convention that needs to be followed for shear.
- If there is a point load acting upwards, then we will take it as positive.
- If it is acting downwards, we will take it as negative.
- If it is to the right side of the section and if it is acting downwards,.
- If it is acting downwards, then we will take it as positive.
- If it is to the left side of the section, and if it is acting upwards, then we will take it as positive.
- Vice versa to this has to be taken as negative.
- If I take a section here and if I take the portion of the beam acting to the right side of the section, if it is anti-clockwise, that moment whatever we are going to act, okay, will bear make the beam bend like a.
- Electrical wire between the two poles, half of it.
- Half of the electrical wire between the two poles.
- If you take only one pole and half of the electrical wire, it is going to bend like this, bend like this, bend like this.
- If the bending moment causes the beam to sag, then we take it as positive.
- You can see that this is like half of the.

Example or application from the lecture

- Then I will consider as I have told you earlier either to the left side or to the right side.
- If there is a point lot that is acting downwards like this, what is indicated, we consider that as positive.
- Whereas if it is opposite to this, that is acting upwards, then we will consider it as negative.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now next coming to this section, if you take a section and if you consider the left side portion of the beam.

Key details

- If the movement is clockwise, it is going to make the beam bend like a sagging, that is like a half moon, okay, or an electrical cable between the two poles.
- If it is on the opposite direction, that is if the movement here, if this movement would have been in the counterclockwise direction, then the beam would have been like this, which is similar to your half rainbow.
- You have cleared with the sign convention what needs to be used for shear force and many movement.
- However, we had three internal effects, but we'll be concentrating only on the shear force and many movement because torsional is not going to be coming here, but you should know what are the three forces that is acting.
- To understand what we need to do in this section is we need to draw the shear force diagram separately across the length of the beam.
- At any point along the length of the beam, if I have the diagram, I can find out what are the values of the shear force and bending movement.
- The design of the beam will be based on the values of the shear force and bending movement.
- The bending movement will actually talk about the longitudinal reinforcement that we need to be provided in a reinforcement concrete beam.
- Whereas the shear force will determine what are the stir-ups that is to be provided in the beam for designing that beam or to carry the load that is coming on that.
- I've chosen a problem 5 r 125.
- The statement of the problem goes like this.
- Determine the shear force and bending movement distribution.
- Distribution means across the length of the beam, what is the distribution of the shear force and bending moment.
- Produced in the beam by the concentrated load.
- And they have also asked you what are the values of the shear force and bending moment at x is equal to L by 2.
- This is what is asked.
- The other end of the beam is free, there is no support, so this is a cantilever beam.
- And there is a point load that is acting at a length of 2l pi 3 from A.
- That is from A to this point where the point load is acting, there is no load.
- And then we can get an equation for shear force and minimum diagram and we can substitute the value.
- of the point from A, whatever is the length, then we can get the shear force amenement diagram or shear force amendment at that point.
- And we can also get one more equation.
- So there will be two equations for this because there is only one point load.
- And substituting the value for in that equation, we will get the value of the sheer monument diagram at a particular point.

- This is a solution for drawing the bending mode and cheap for diagram.
- First thing that we need to do is we have to draw a free body diagram.
- Free body diagram is free of all the supports.
- The supports have been removed and replaced it by the respective reactions.
- Then we have the point load, distance is also mentioned, the beam is mentioned with respect to a horizontal line because the depth of the beam is not going to affect the shear force embedding moment.
- Now we need to first find out the.
- Support reactions for which we will use the equations of equilibrium.
- This is very well known to all of you.
- You have already solved several problems on external effects of the beam.

Example or application from the lecture

- So let's take a simple example to understand how do we draw the shear force and diagram.
- Now, if you read the diagram that is given, this is a cantilever beam because you have a fixed support at A.
- To draw the shear force unbending moment diagram, we should consider a section in the portion where there is no variation in the load.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 15:00–19:59

Horizontal reaction is zero, vertical reaction is P.

Key details

- Moment we will get as 2PL by 3, which will be acting in the counterclockwise direction.
- Diagram or the bending moment.
- I will take a section here that is X.
- The distance from A to this section is considered as X itself.
- Section the beam or to the right side, if you take it to the left side, what we get for shear forces we have VA which is equal to P, that is what we have got it from here.
- Then if I take the movement MA, it is acting in the counterclockwise direction.
- So substituting the value of MAR and substituting the value of V A, so it becomes P into X minus because this MA, we know that it is Augging, so it is negative 2 P L by 3.
- This is the equation for shear force that is VA equals P, and the bending moment equation is P x.
- Minus 2PL by 3 is the equation for many limited.
- We also have this point load P.
- In this zone, that is from the point where the point load is acting to the end of the beam, shear force is zero.
- Next to take the bending moment, again we will take the moment, so we will get VA into hex distance.

- You have the moment MA which will be acting or making the beam to bend like a hogging, so it is negative, whereas VA is sagging, make the beam to bend like sagging, so it is positive.
- Using this, we can draw the shear force and bending movement diagram.
- But however, in the problem they have asked you to find out.
- The value of the shear force admitting moment at a point where x is equal to L by 2.
- Simplifying that we will get the value for the shear force it is.
- Minus P L by 6.
- This is the value that we are going to get.
- I hope all of you have understood how we can get the internal effects or what are the internal effects that is acting on a beam.
- How or what is the sign convention that should be used for shear force and bending movement diagram.
- Also, you have understood how do we get the distribution of the shear force and bending moment diagram between the zones or the points where there is no variation in the load, and how to get the value of the shear force and bending moment bending moment from the distribution.
- We'll be continuing in the next class.
- This was class number 53.
- In the next class, we will be solving a few more problems on how do we draw the bending moment and GFO diagram and how do we determine the shear force and bending movement diagram or shear force and bending movement values at a particular point.
- My name is Dr.
- I am from the civil engineering department of science and humanities PS University.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

11 UE25CV141A_Unit 3_Class 54_AV Summary

Study focus: UE25CV141A_Unit 3_Class 54_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SUV Vinkitish from Civil Engineering under the Department of Science and Humanities, PES University.
- This is class number 54.

- We are in Unit three, Distributed Forces.
- And chapter is Distributed Forces.
- Topic is beams internal effect.
- In the last class, that is class number 53, we had discussed about a problem that is phi bar 125.
- So first problem that I've chosen for today's class is phi bar 126.
- The statement of the problem goes like this.
- Determine the shear force V at a section B between A and C.
- And the bending movement at the support A.
- The point load is acting at the free end of the beam of magnitude 1.4 kilonewton and it is acting downwards as you can see.
- Then we have one more point load, which is acting at a distance of 1.5 meters from the free end, that is from the left end.
- of magnitude 1.8 kilonewtons.
- Then we have a roller support at point A, which is at a distance of 1.6 meters from the lower point 1.8 kilonewtons.
- And you have one more support at point C.
- Which is at a distance of 1.8 meters from support A.
- If you observe the support A, as I have told you, it is a roller support, and the support at C is a hinge or a pin support.
- So, first thing that we have to do here is to determine the support reactions as we have to determine the shear force at B.
- Whether we take to the left side of the section or to the right side of the section, B, we need to determine the support reaction at A.
- And C to find out or to take the section to the right side of the beam.
- So to first to find out the support reaction, as we have explained in beam's internal effect, we have to use the three equations of equilibrium.
- The free body diagram looks like this.
- What do we have in the free body diagram is the point load of one point four kilometers is acting downwards at the free end, that is the left side end.
- Then one more load that is of one point eight kilonewtons, which is acting at a distance of one point five meters from the left end.
- Then we have the support at A which is being replaced by its reaction R A, since it is a roller support, the reaction will be only one and it is acting upwards and it is denoted as RA.
- Then we have a support at C which is a pin support or a hint support, so we will have two reactions, one vertical, which is denoted as V C and one horizontal, which is denoted as H C.
- For which we will use the equations of equilibrium.
- So we will take first equation of equilibrium as summation of the moment to be equal to 0, and I have taken the moment at point C because we have two reactions acting at this point.
- We get this point load.
- Which is 1.4 kilometer into the perpendicular distance is totally equal to 1.5 plus 1.6 plus 1.8.
- That is what is indicated here, or you can take it from C, which is 1.8 plus 1.6 plus 1.5, and that's what is indicated here.

- Point load that is one point four kilometer will produce an anti clockwise movement, so we will take it as positive.
- Which is also a positive because it is acting in the anti-clockwise direction.
- Then we have one more point load that is acting because of the support at A, which is upwards, which will create a clockwise moment at C.

Example or application from the lecture

- The diagram, if you read the diagram, what is given in the diagram is there is a point load that is acting.
- So first let us draw the free body diagram.
- So first let's find out the values of RA, V C and H C.

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EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

And that is what we have the forces acting on this beam.

Key details

- Next, to find out the other support that is V C, we will take all the summation of all the forces, algebraic sum of all the forces acting along the y direction to be equal to 0.
- But one R A which is equal to 7.2 kilon, which is acting upwards, which will be positive.
- Which is equal to minus 4.01 kilonewt, which indicates that since there is a negative sign here.
- That indicates that the reaction at C is not acting upwards, it is acting downwards.
- And that is possible because we have a support at C as an heat support, so we can have either upwards reaction or downwards reaction.
- I hope that is clear.
- Next we will take the other equation of equilibrium that is algebraic sum of all the forces acting along the x direction to be equal to 0.
- So we will get HA to be equal to 0 because there is no horizontal force acting on the beam, so H will be equal to 0.
- What is asked in the problem is to determine the shear force at a section B, that is what is indicated here, which is just to the right side of A.
- However, here in this problem I have shown you by both, whether you take to the left or to take to the right, using the sign convention that was mentioned in the last class, you will get the same value for the shear force.
- That is what I've indicated and that is why.
- You can see the free body diagram that is written here, where I have taken only the beam acting to the left side of the section beam.
- This is what is the beam body diagram.
- We have only VC and what we have got for V C is negative, which indicates that it is acting downwards, so that is what is indicated here.
- Next, they have also asked us to determine the movement about A.

- Similar to the shear force, I have also I have written the diagram of the beam.
- Which is to the left side of A and right side of A, that is this one.
- We have point load one point four and one point eight, both of them acting downwards.
- And at A we also have a reaction at A R A.
- Then if you take to the right side of the section, again we have a point load or the reaction at V C that is acting, which is downwards, which is four point zero one, and the reaction at A that is acting.
- Going forward to find out the shear force at V here, if I take the section to be acting to the.
- Left side or the beams acting to the left side, what do we get?
- We get 1.4 acting downwards, which is negative.
- 1.8 is also acting downwards to the left side of the section, so it is negative.
- Then we have.
- RA which is acting to the left side of the section upwards which is positive.

Example or application from the lecture

- When we consider to the right side of the section.
- Whatever value that we get, that should be the same value what we get it when we consider to the right side of the section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 10:00–14:59

Let's see that.

Key details

- Then coming to the bending moment, we take the bending movement at this point.
- Whatever I have discussed, the beam taken to the left side of the section B.
- Shear force at B will be equal to 1.8, which is acting downwards.
- Then we have 1.8 which is acting downwards, 1.4 is also acting downwards, both of them is indicated here.
- Plus RA which is acting upwards, so that is 7.21.
- Considering to the right side of the section, that is this one, we get the beam acting to the right side and the point rod is acting downwards.
- Next, to find out the bending moment, similarly, take the section to the left side from A.
- 1.6 is the distance, that is this distance, plus 1.5, so that is what we have written here.
- one point eight, which will also be hanging.
- into 1.6 so we get the value as 7.22.
- This is with respect to when we take the beam to the left side of the section.
- We get 4.01 into this distance of 1.8 which will also be Augin.
- Minus 7.219.
- That is what I have proved here for shear force as well as petting movement.
- I hope you have understood this problem.
- The problem statements go like this.

- Draw the shear and bending movement diagrams for the diving board, which supports the ATKG man poised to dive.
- Specify the bending moment with the maximum magnitude.
- In the diagram, the man is acting at the free end of the board.
- And also we should specify the maximum winning moment.
- And also the distance between A and B is one point two meters, whereas the distance between A to the free end of the beam or the board where the person is standing is four meters.
- So, first we have to find out the support reactions for which we will take the equations of equilibrium, considering summation of all the moments acting at A to be equal to 0.
- We will get when we take moment about A, we will get the weight of the person, which is 18 to 9.81, and the distance from A to the point where the mo where the person is.
- Is acting or the force of the person is acting is 4 meters, so that is 4 here.
- That would produce a clockwise movement.

Example or application from the lecture

- Next, let's go to the next problem, which is 5 bar 128.
- The second convention used is given here, clockwise is positive.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then we will take summation of all the forces acting upwards to be equal to 0.

Key details

- Of course it is indicated that the reaction is acting downwards, but what is taken here is that it is acting upwards, that is why it is positive, which is equal to zero.
- But once we calculate the value, we get R1 to be equal to negative, and that is why this is indicated as downwards.
- Remember that this R1 is positive because we are assuming it to be acting upwards.
- What we can do is.
- You write down horizontal lines under the point loads plus the support reactions, that's what is indicated here.
- Then draw a datum, draw a datum line, which indicates a 0, 0.
- You go down as the arrow of the support reaction is indicated, which is downwards of R1, then there is no load, so horizontal line.
- Then at B we have a reaction acting upwards, which is 26.
- You take it from the horizontal line above, so you get a value of 785.
- That is 2650 minus 1875.
- The point where the point load is acting to the edge of the beam, there is no load, so horizontal then go down by H and 5.
- For bending moment, determine the moment at B either taking the section to the left side or right side.

- And then draw the bending moment diagram.
- I hope all of you have understood.
- Now next we have asked you to find out the maximum bending moment in the beam, which will occur at the point where the support is acting at B that is equal to the R1 into the distance of.
- A to B.
- I hope all of you have understood this.
- We are going to be stopping this class here.
- My name is Dr.
- Sv Mektesh from civil engineering and the department of.
- Science of Minnesota, PS University.
- See you in the next class.
- That is class number 55.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

12 UE25CV141A_Unit 3_Class 55_AV Summary

Study focus: UE25CV141A_Unit 3_Class 55_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Civil Engineering from the Department of Science and Humanities, PS University.
- This is class number fifty-five.
- In the last class, that is class number fifty-four, we had discussed about how to determine the shear force and bending movement for a beam which is loaded with different types of loadings.
- We will continue.
- the same topic in this class also, and we will see how to solve some more problems which are loaded with different types of loadings.
- And we are dealing with the topic under unit three, that is distributed forces, and in that chapter five, and the topic is beams internal effects.
- And we have referred to the textbook, Engineer Mechanics, Statics, 8th edition, the authors are Miriam Craig and Bolton.
- For today's class I have chosen this problem that is problem number 5 bar 127.
- The statement of the problem goes like this.

- Draw the shear and bending moment diagrams for the beam subjected to the end couple.
- What is the bending moment M at a section 0.5 meter to the right of beam.
- which is subjected to a couple at the end of the beam, that is to the right side of the beam.
- And also we need to find out what would be the value of the pending moment.
- At a section 0.5 meters from the right side of the support B.
- Which is supported at point A by a pin or a hinge support and at B by a roller support.
- The distance between A and B is 2 meters.
- At the end of the right side of the beam,.
- There is a couple or a moment that is acting in the clockwise direction of magnitude 120 newton meters.
- As well as we need to draw the bending movement diagram for the beam.
- After drawing the shear force and bending moment diagram, we are supposed to find out what is the bending moment acting at a point right side of B at a distance of 0.5 meters from B.
- That is this distance, if you take a section here.
- This distance has to be 0.5 meters and we need to find out what will be the bending moment at that point.
- And it is the beam is supported at A by mean support under log support.
- First thing that we need to do to solve this problem is to determine the external support reactions acting at A and acting at B.
- We will first draw the free body diagram by replacing the supports by their respective reactions.
- Then there is an horizontal reaction acting at A, we will assume it to be acting to the right.
- Then at B point we have a roller support.
- Reaction at B is upwards only.
- Whereas the sense or the reaction at A can be upwards or downwards, as well as it can be to the right or to the left because it's an hint support.
- When I take the algebraic sum of all the forces acting along the x axis that is horizontal and equality to zero, then I will get the reaction at A along the horizontal which should be equal to zero because there is no horizontal force that is acting.

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EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then substituting the values between this.

Key details

- section that is between A and B and B and C.
- We can get the different values for shear force and body moment at any particular point between A and B and B and end of the V.
- That's what is done here.
- We have I have written the free body diagram here.
- The hint support is replaced by its reaction, vertical reaction acting upwards.

- I have assumed it to be VA and horizontal reaction acting at A, I have assumed it to be HA.
- Then the reaction at B which is the because of the roller support and it will be acting upwards.
- Then there is a movement acting at the free end of the beam, which is 120 newton meters, and that is what is shown.
- Equilibrium that is used R sigma M at A equals 0, as I have explained.
- Using that I have determined what should be the reaction that is going to be acting at B, which we will get it as 60 kilonewton.
- Then using the algebraic sum of all the forces acting.
- along the vertical, we will find out what will be the reaction at A and we will find that the reaction at A is acting downwards BA.
- Then we will also find out what is the horizontal reaction at A by taking all the forces along X, but we found it to be 0.
- Next, we will take a section in between A and B, okay, and that distance from A to be equal to X, that is what we will take.
- We will take all the forces acting to the left side of the section and we will determine the shear force equation.
- All the forces acting to the left, what we have is only the reaction acting at A, that is VA.
- Which is equal to minus R B, which is equal to 60 kilonot minus R B and it is acting downwards.
- To the left and downwards, you will get it as negative minus 60, and that's what is indicated here.
- And that value will remain constant for the entire section between A and B.
- Next, if you take a section B C here, okay, any section the distance from B is X, if you take what will be the shear force.
- The values are same, so we will get the shear force to be equal to zero.
- Using these two equations we can draw the shear force diagram as indicated here.
- Between A and B, it's a constant value and it is minus 60, so that is what is indicated.
- This is the datum line 00.
- Then we have a 60 between this, which is a constant value, and then between the section B and the end of the beam.
- it is the shear force is zero, so we have we have drawn that shear force diagram.
- We will take a section here.
- This distance is X.
- Then we will write what will be the bending moment.
- VA I have assumed it to be acting upwards.
- plus 60 into x, so we will get it as negative minus 60 into x because that will produce a hawking bending moment.
- Similarly, if I take a section between B and C, whereas this is C and this is B, this is X distance, and we will take all the moments acting or all the forces acting to the left of the beam and taking movement of all these forces, we will get bending moment to be equal to minus 120, that is because.
- we will have this 60 into this distance that is 2 plus x, whereas this 60 into x we will get.
- Or if I take the section to the right side of the beam.

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EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

We will get a movement which is equal to 120, which will produce a hawking bending movement, so it is minus 120.

Key details

- But the support A and the free end it is 00.
- As we progress from A to B, as X increases, your bending moment also increases by 60 times X.
- And between B and C that is N of the beam, it's a constant value which is one twenty and it is negative, so it will get a constant value.
- So drawing this, we will get this diagram and this is negative because it is below the datum line and this is the bending moment diagram.
- The next part of the problem is to determine what would be the bending moment value at a distance 0.5 meters from B to the right side, that is at this, this distance is 0.5.
- that the bending moment is a constant value which is 120 that is negative bending moment.
- However, however we can get that.
- Then the shear force diagram and bending moment diagram together.
- We have determined the support reactions, which we have got it as 60 and 60 here, but at E it is acting downwards, at B it is acting upwards.
- Bending moment at point five meters, we will get it to be one twenty, that is what is indicated here.
- I hope all of you have understood this.
- We will move on and to the next.
- Question I have considered a question 59.
- Draw the shear and bending or draw the shear and movement diagrams for the loaded breed.
- What are the values of the shear force and bending moment at the middle of the beam is what is ask.
- And after that, we have to determine what would be the value of the shear force and bending moment at the middle of the beam.
- It's a simply supported beam.
- At the left hand, that is at A, we have an inch support or a pin support.
- At the right hand, B we have a roller support.
- Then we have a point load acting downwards of magnitude 4 kiloneeton at a distance of 1 meter from A, which we will take it as point C.
- Then we also have a counterclockwise moment of magnitude 2800 newton meters acting at a this point, which we will take it as D, which is at a distance of 2 meters from A, whereas 1 meter from C and as well as 1 meter from B to the left side.
- So, again in this problem, if I want to draw the shear force on bending moment diagram, first thing that we need to do is to draw or to draw the free body diagram and then to determine the support reaction's value.

- Hinch so two reaction, one vertical and one horizontal, but B there is roller, always it is going to be acting upwards because the roller as you can see it is indicated like this.
- I have replaced the.
- Heat support by two reactions, vertical and horizontal, then roller support at B by one reaction RB and I have indicated the load that is acting on the beam.
- Then first equations of equivalent that I have taken is sigma fx is equal to 0, with which I will get movement or the horizontal force at A to be equal to 0.
- Then the next equation taken as algebraic sum of all the vertical forces to be equal to 0.
- 4 kilonut which I have converted into newtons which is 4000 acting downwards, so negative.
- Then we have R B acting upwards or positive, R B equals 0.
- So what we will get is we will get 4000, that is because of this point load into 1 meter.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

Which is clockwise, so negative.

Key details

- Then we have this movement which is anti-clockwise, so 2800 is acting in anticlockwise direction, so positive.
- Then R B will also produce a counterclockwise movement, so that is also positive.
- From this equation we will get the value of R B which is equal to 400 newtons.
- We will substitute this 400 newtons in this equation and get the value of VA, which will be equal to 3600 newtons.
- Reaction at A vertical reaction is 3600 newtons.
- Vertical reaction at B is 400 newtons.
- And both the reactions at A and B which are vertical are acting upwards itself.
- So shear force is equal to Va because there is only one reaction that is acting to the left of the beam that is VA, so that's a constant value which we will get as 3600 as indicated here.
- Next, we will take a section between C and D, that is here.
- If you take the left side of the section, we will get shear force equals 400, that is VA, which is acting upwards, 3600, and 4000 newtons, which is acting downwards, so negative.
- between the section C and D.
- From A is X that we have taken.
- Minus 4000 into X minus 1.
- Because I have taken the X distance from A to this point as X, it is.
- X minus 1, which will be this distance.
- This is the datum line 00.
- Between this section, it's a constant value of 3600.
- Then coming to this section, again, shear force is a constant value which is minus 400.
- Then till the.
- C to B, it is constant itself because there is no vertical force acting between this.
- This is a shear force diagram.

- Above datum line is positive, below datum line is negative.
- The bending movement value is also increased, and if I put x is equal to 1, I get it as 3600, that is a point B, so it is a linear line because it's a point load.
- Then between C and D.
- Also we have this equation substituting the value of x, anything more than one and less than two, we will get the variation at x is equal to zero, we get it as 3600 and when x is equal to 2 we get the value of 3200.
- Then considering the moment we will get it to be 400 itself, and that is what we have indicated here.
- When we take the section B and D we get it as 400 when X is equal to 0 we will get it this to be 400 and that is what is indicated and when x is equal to 0 we will get x to be 0 when x is equal to 1 meter because we have taken it to the right side we will get this to be 4.
- Then he has asked you to determine what will be the value of the shear force embedding moment at the middle of the beam, which will be at a distance of 1.5 meters from A as well as 1.5 meters from B.
- From the shear force embeddement moment, if I take it.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 20:00–24:59

I the shear force value is four hundred, whereas the bending moment value we have to compute.

Key details

- It is varies from three thousand six hundred to three thousand two hundred.
- Or 3600 minus 3200, which is 400, and since it is the middle, it will be 200 mm, which is lesser than the value at B and 200 m 200 newton millimeters greater than the value at C.
- I'll explain this clearly.
- My name is Dr.
- Svintesh from Civil Engineering under the Department of Science and Humanities PS University.
- Discuss in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

13 UE26ME142A_Unit 3_Class 44.1_AV Summary

Study focus: UE26ME142A_Unit 3_Class 44.1_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkitesh, a civil engineer from the Department of Science and Humanities, PS University.
- This is class number 56.
- We'll be continuing to solve few more problems in today's class.
- And we are discussing about the.
- Unit 3 distributed forces in that chapter 5, beams internal effect, and we are referring to the engineer mechanics statics book written by Miriam and Craig, and it is the 8th edition.
- The problem number is 5 bar 130.
- The statement of the problem goes like this.
- Determine the shear force V and bending moment M at a section of the loaded beam 200 mm to the right of A.
- It's a simply supported beam.
- At the left hand, that is at the point A, you have a hint support or a pin connection or pin support, and then at the right end, that is at point B, we have a roller support.
- And to this beam, there is an uniformly distributed load or a rectangle loading of intensity 6 kN per meter for a distance of 300 mm from the left support that is from A.
- The length of the load is three hundred mm, whereas the total length of the beam is three hundred mm plus three hundred mm, totally it is six hundred mm.
- 200 mm will be somewhere here.
- So if I want to determine the shear force and bending moment, first thing that I need to do is to find out what is the support reactions.
- To determine the support reactions.
- I have to draw the free body diagram.
- To draw the free body diagram I have to replace the support by its respective supports.
- That is one vertical reaction at A and one horizontal reaction, vertical reaction BA and horizontal reaction H A.
- Then at B I will replace this by one reaction that is R B which is acting vertically upwards.
- Then the uniformly distributed load will be replaced by a point load of magnitude equal to the.
- So once I draw this free body diagram, next I'll apply the equations of equilibrium, all the three equations of equilibrium, to determine the vertical reaction at A, horizontal reaction at A, and then vertical reaction at B.
- Then reaction at B by R B and then the UDL by a point load of magnitude 1.8 kilonewtons, which is acting at a distance of 0.15 meters from A.
- I get V A into this total distance, which is 0.6 meters, which will produce a clockwise movement.
- Whereas this one would produce an anti-clockwise moment, so it is minus 1.8 into the distance from this to this is 0.6 minus 0.15, which is 0.14, that's what is mentioned here.

Example or application from the lecture

- And the diagram that is given here is the diagram that is given.
- Length of the load that is 0.3 mm because the intensity of the loading is given as 6 kN per meter to have consistent in units, I'll convert this distance 300 mm into 0.3 meters.
- Once I get these reactions, then I can consider the section at 200 mm from a I can take either left side of the section or the light side of the section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

We will get V A to be equal to 1.35 kilon and we got a positive value.

Key details

- So that means the sense of VA is acting upwards.
- Next, once I get this, of course I can determine the reaction at B by using.
- The algebraic sum of all the vertical forces acting to be equal to zero.
- But however, since I have to determine the shear force and bearing moment at 200 mm from A, I need not.
- I have to determine RB.
- Instead I can take the section of the beam to the left side and then solve it easily.
- So that's why I have not determined the reaction at B or the reaction at HA of obviously H A is going to be zero because there is no horizontal force that is acting.
- The section 200 mm.
- What do we have?
- We have VA which is acting vertically upwards.
- Then very importantly, what we need to draw here is whatever is the load that is acting till 200 mm is what we need to take.
- UDL which is acting of my intensity 6 kN per meter and for a distance of 0.2 meters.
- The point load was 6 into 0.3.
- It's the total load that we are taking, whatever is the load that is acting on the beam.
- But when we are taking a section to find out the shear force embedding moment, which is at point two meters from A, we have to take only that load.
- I hope this is very clear.
- The force is acting upward that is V A, so it is positive, which has a magnitude of one point three five kilonewton.
- Then to the left of the section and downwards, so it would be negative of the value 1.2, so that is what is mentioned here.
- That is the shear force that is acting at the section 200 mm to the right of A.
- Similarly to determine the bending movement, so bending movement at this point will be equal to V VA into point two, which is the moment, and that causes a sagging bending movement because the beam is going to bend like this.
- Then because of this load, the beam is going to bend like this, which is a hogging bending movement.

- The distance 0.1, which is mentioned here.
- support A and that's what we have done.
- You can see the free body diagram of the beam.
- The UDL has been converted into a point load and represented as R.
- Taking moment about B we get RA, which is a value 1.35 kilonewton.
- Where we want the shear force of minimum that is 200 mm from A.
- This is the diagram.
- The sum of all the forces acting unequality to zero, so you get the shear V is equal to 0.15 kilonaut time, then takes algebraic sum of all the movements acting at A to be equal to 0.

Example or application from the lecture

- Earlier when we consider the entire beam.
- What is the value of R is given here, that is 6 into 0.3, which is 1.8 kilonewton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

But the section 200 mm from A, which is equal to 0.15 kilonewton per meter.

Key details

- I hope all of you have understood this problem.
- So next we will move on and this is the next problem that I have considered.
- Problem number 5 bar 132.
- The problem statement goes like this.
- Determine the shear force V and bending moment M in the beam at a section 2 meter to the right of end A.
- The beam was subjected to a UDL, whereas in this problem, the beam is a simply supported beam again, of span 6 meters.
- The left support is a hinge support, the right support is a roller support.
- But here the loading is a uniformly varying load or a triangular load with zero intensity at point A and maximum intensity of 4.8 kN per meter at a distance of 3 meters from A.
- Once you get the reaction at A.
- Draw a separate diagram.
- I have drawn the free body diagram of the beam, replacing the support by its reaction B A and H A at point A where there was hinge.
- Then the triangular loading.
- The magnitude of that would be half of 3, which is the span of the loading into the intensity of loading, which is 4.8.
- The point of application of this is going to be at a distance of two thirds from A.
- Then at point P the roller support is replaced by a reaction R B, vertical reaction R B.
- Four point eight kilonewtons what we have got.
- Next we will draw only the.
- Forces that is acting at 2 meters from V A.

- Considering the left side of the support, so this is what we get.
- We get a vertical reaction VA, which is of 4.8 kilonewtons.
- For 2 meters a triangular loading.
- When we have to convert this triangular loading into a point load, we should find out what is the intensity of the load at two meters from A.
- This is one triangle, and then this is another triangle, they are similar.
- of 2 from A, which I have mentioned it here.
- 2 thirds of 2 that is 4 by 3.
- But section 2 meters from the right of End A, shear force will be we have VA to the left of the section and is acting upwards, so it is positive, whereas 3.2 is to the left and downwards, it is negative.
- Coming to bending moment, considering the moment at two meters from point A.
- We have VA into A which will cause a saging bending moment, so it's positive, whereas 3.2 will cause an hogging bending movement, so it is negative, so it's 3.2 into the distances 2 by 3.

Example or application from the lecture

- Consider a section 2 meters from A.
- Consider all the forces that is acting in this 2 meter span of the beam, and then find out the shear force and bending movement at that point.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

Sorry, I have left out the kilonewton meters here.

Key details

- Please kindly make that correction.
- Replacing the triangular loading by a point load, then determining what is the reaction at RA, which is 4.8 kilon, then considering only the portion of the beam to the left side of the section that is at 2 meters from here.
- Whatever is the loading that is acting within this pan, that is for two meters, this is a triangle.
- We have EA acting here.
- Then determining the distance at which this R1 is acting, which he has considered as A.
- Which is 2 thirds of 2, which is 4 by 3 meters.
- Then R1 is half of 3.2 into 2, that is 3.2 kilonewtons.
- Using this, we can find out the shear force considering all the vertical forces acting on the beam.
- Then taking movement about this point and equating it to 0, we get.
- Minus 3.2 into 4 by 3 which is this force.
- Then minus one point six into two.
- We are equal to zero.

- We are equal to zero.
- I hope everybody is clear with this.
- If you have any doubts, you can always contact me.
- I am Dr.
- S V Vength, Civil Engineering from Department of Science and Humanities, PS University.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

14 Source slide deck: Centroid

Study focus: Source slide deck: Centroid. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000202 · EV0000203

SHORT-NOTE RECAP

Study notes · Source slide deck: Centroid

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

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Slide 2 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS

- 5/47 Determine the coordinates of the centroid of the trapezoidal area shown.
- =A=2400(30)+600(20)

Slide 4 · Slide page 4

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

Slide 5 · ENGINEERING MECHANICS

- 5/48 Determine the distance H from the upper surface of the symmetric double-T
- beam cross section to the location of the centroid.
- 20015)(# + 2(1520)(5+2)

Slide 6 · Slide page 6

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

Slide 7 · ENGINEERING MECHANICS

- 5/49 Determine the x- and y-coordinates of the centroid of the shaded area.
- 5/50 Determine the height above the base of the centroid
- 5/51 Determine the x-and y-coordinates of the centroid of

Slide 8 · Slide page 8

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Slide 9 · Slide page 9

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Slide 18 · Slide page 18

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Slide 19 · Slide page 19

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Slide 20 · Slide page 20

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Slide 21 · P. Ramchandra

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EV0000202 · EV0000203 · EV0000204 · +30 source units in pack.json

15 Source slide deck: UE26ME142A Unit 3 Class 33 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 3 Class 33 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000235 · EV0000236

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 3 Class 33 Lecture PPT

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Slide 1 · Slide page 1

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Slide 2 · Engineering Mechanics

- J. L. Meriam and L. G. Kraige
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- S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces

Slide 3 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 4 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 5 · ENGINEERING MECHANICS STATICS

- Determine the Centroid for an “area of a circular sector”.
- Solution II. The area may also be covered by swinging a
- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 6 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 7 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 8 · ENGINEERING MECHANICS STATICS

- rmine the x and y - coordinates of the centroid of the shaded area.
- 66.74(103) – 50.3(103) =
- S V Venkatesh - Engineering Mechanics Statics/ 4740– Unit 3 – Chapter 5 – Distributed Forces

Slide 9 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 10 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 11 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 12 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 13 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 14 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 15 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 16 · Slide page 16

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EV0000235 · EV0000236 · EV0000237 · +41 source units in pack.json

16 Source slide deck: UE20CV101 UNIT 4 SESSION 2

Study focus: Source slide deck: UE20CV101 UNIT 4 SESSION 2. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000279 · EV0000280

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 4 SESSION 2

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Slide 1 · ENGINEERING MECHANICS

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Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Centroid of the area of a circular sector:
- Consider an element at a distance $ro$from the centre O of a sector,
- radial width being dro& bound by radii at θ& (θ+ $d\theta$)

Slide 4 · ENGINEERING MECHANICS - STATICS

- Locate the centroid of the area of a semicircle
- Consider an element at a distance $ro$from the centre O
- of a semicircle, radial width being dro& bound by radii

Slide 5 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 6 · ENGINEERING MECHANICS - STATICS

- ⊔y= b h h h 3 , ⊔y= ⊓x= C C 3 3
- h h C ⊓x= $-2b$, ⊔y= h ⊓x= $-2b$, ⊔y= $-2h$ 3 3

Slide 7 · ENGINEERING MECHANICS - STATICS

- C C x ⊓x= 0, ⊔y= 0

Slide 8 · ENGINEERING MECHANICS - STATICS

- C x O R C
- ⊓x= R, ⊔y= $-R-4R$ ⊓x= $-R-4R$ ⊔y= 0 3π 3π,

Slide 9 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 10 · ENGINEERING MECHANICS - STATICS

- The area is divided into strips of width dx and variable
- height h. The area A of each strip, such as the one shown
- in red, is h dx and is multiplied by the coordinates $xc$and

Slide 11 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000279 · EV0000280 · EV0000281 · +30 source units in pack.json

17 Source slide deck: UE24CV141A Unit 3 Beams

Study focus: Source slide deck: UE24CV141A Unit 3 Beams. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000366 · EV0000367

SHORT-NOTE RECAP

Study notes · Source slide deck: UE24CV141A Unit 3 Beams

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Slide 2 · Engineering Mechanics

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Slide 3 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces
- S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces

Slide 4 · Slide page 4

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Slide 5 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 6 · Slide page 6

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EV0000366 · EV0000367 · EV0000368 · +9 source units in pack.json

18 Source slide deck: UE25CV141A Unit 3 Class 48 Lecture PPT

Study focus: Source slide deck: UE25CV141A Unit 3 Class 48 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000378 · EV0000379

SHORT-NOTE RECAP

Study notes · Source slide deck: UE25CV141A Unit 3 Class 48 Lecture PPT

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Slide 1 · Slide page 1

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Slide 2 · Engineering Mechanics

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- S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces

Slide 3 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces
- S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces

Slide 4 · Slide page 4

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Slide 5 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Slide 6 · Slide page 6

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EV0000378 · EV0000379 · EV0000380 · +9 source units in pack.json

19 Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 8

Study focus: Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 8. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000390 · EV0000391

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 8

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS

- 5/101 Determine the reactions at A and B for the beam
- subjected to the uniform load distribution.

Slide 4 · ENGINEERING MECHANICS

- R= 6 X (300/1000)= 1.8 KN
- Ax = 0 FBD of beam AB

Slide 5 · ENGINEERING MECHANICS

- 5/103. Calculate the reactions at A and B for the beam

Slide 6 · ENGINEERING MECHANICS

- R1= w. l = 400 X (8)= 3.2 KN
- 1 R2 = w. l/2 = 400 X (6)/2= 1.2 KN
- l/3 w.r.t A = 6/2= 2m

Slide 7 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000390 · EV0000391 · EV0000392 · +18 source units in pack.json

20 Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 9

Study focus: Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 9. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000411 · EV0000412

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 9

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

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Slide 2 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS

- 5/105. Find the reaction at A due to the uniform loading

Slide 4 · ENGINEERING MECHANICS

- MA= 3 KN.m CW

Slide 5 · ENGINEERING MECHANICS

- 5/116. For the beam and loading shown, determine the
- magnitude of the force F for which the vertical reactions at A
- and B are equal. With this value of F, compute the magnitude

Slide 6 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 7 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000411 · EV0000412 · EV0000413 · +18 source units in pack.json

21 Source slide deck: UE26ME142A UNIT 3 ONLINE SESSION 41

Study focus: Source slide deck: UE26ME142A UNIT 3 ONLINE SESSION 41. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000432 · EV0000433

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A UNIT 3 ONLINE SESSION 41

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Slide 1 · ENGINEERING MECHANICS

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Slide 2 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS

- 5/107 Determine the reactions at A and B for the beam

Slide 4 · ENGINEERING MECHANICS

- R= 18 X (4)= 72 KN
- Ax = 0 FBD of beam AB
- 15 + (RB x 4) - (72 x (2+2+2)) = 0 72KN

Slide 5 · ENGINEERING MECHANICS

- 5/109. Determine the reactions at A, B, and D for the pair of
- beams connected by the ideal pin at C and subjected to the

Slide 6 · ENGINEERING MECHANICS

- AX = 0 BY = 29.4KN
- AY = 12.96KN DY = 2.67KN

Slide 7 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000432 · EV0000433 · EV0000434 · +18 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
CLASS 33 Introduction to Centroid transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 35 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 29 Introduction to Beams transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 31 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE25CV141A_Unit 3_Class 53_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE25CV141A_Unit 3_Class 54_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Centroid	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Centroid	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Centroid (converted PDF; original hash 6f51d7bc8a1daab8830750f6f5293f7b5339619f4152cedcb407c7e55a2b64f5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 3 Class 33 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A Unit 3 Class 33 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 3 Class 33 Lecture PPT (converted PDF; original hash 21e5fd6df4dee659c3d04ed9736319637d645f4506f5cdc7a06008eb470938d4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 2 (converted PDF; original hash 2637a5f5acfd6be279cbdc1ada5c369beec1a971e612312a6303ceb42b71ded8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 3 (converted PDF; original hash 855d90c138ebd09ccd87444c3ca92bc80298d14585d21a759dab47f34f50f340)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 4 (converted PDF; original hash 98aff97f5c1596c1d7d2a9a246bd2617a83cb4eb8293cdac8a7732f46224d2e5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CV141A Unit 3 Beams	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE24CV141A Unit 3 Beams	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE24CV141A Unit 3 Beams (converted PDF; original hash ebb61d8de2a89dc7df0dd92e8f1b503458f063b51cbf37478c9aebffc2651e41)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 48 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 48 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 48 Lecture PPT (converted PDF; original hash ebb61d8de2a89dc7df0dd92e8f1b503458f063b51cbf37478c9aebffc2651e41)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 3 ONLINE SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE20CV101 UNIT 3 ONLINE SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 3 ONLINE SESSION 8 (converted PDF; original hash 1ba5fc810a2d6b1882d2198242bbf43c152b21e3b65ae248aa8423782827ef48)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 3 ONLINE SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 3 ONLINE SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 3 ONLINE SESSION 9 (converted PDF; original hash 0b4358b69167fcc42fb2884667d9299882cabd80425a2a68535a12554f7ac094)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A UNIT 3 ONLINE SESSION 41	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A UNIT 3 ONLINE SESSION 41	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A UNIT 3 ONLINE SESSION 41 (converted PDF; original hash d50b5f40928a41599890eee7efc577bfe0f2f9c52818f0c722a071108930895e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 53 Lecture PPT (converted PDF; original hash 764f7c39a17dfafc7dee861f13a669ecec990a216b3bc40402cf6930874a3d6b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 54 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE25CV141A Unit 3 Class 54 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 54 Lecture PPT (converted PDF; original hash a2b0aaf2ea90e653a4f79c8bbbf71b30fabe0cca15e221d3cdbee88b44728b9d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 55 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 55 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 55 Lecture PPT (converted PDF; original hash ab718cce2de3498ea99fc0ee34a3028cc61a5fd54dd151458ff8a2936eac7ef2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 3 Class 44.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A Unit 3 Class 44.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 3 Class 44.1 Lecture PPT (converted PDF; original hash 2ef2625de901226124ab0b819e8dc0e27969ca67ff246d04970156ac6728792a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.

9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00042	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00044	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00050	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00052	21 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00051	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00053	14 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00055	16 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00054	14 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00056	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00058	11 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00057	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00059	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00061	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00060	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00062	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00064	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00063	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00065	3 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00067	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00066	3 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00068	3 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00070	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00069	3 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00071	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00073	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00072	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00074	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00076	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00075	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00077	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00079	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00078	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00080	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00082	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00081	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00083	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00085	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00084	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00086	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00088	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00087	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00089	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00091	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00090	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

C Revision and retrieval practice